# Dune Collisions in 2D

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# 1 Motivation and Model Background

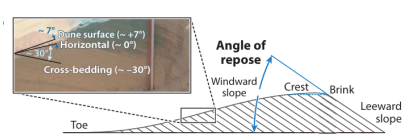
## 1.1 Introduction

Dunes appear to migrate across a desert landscape as cohesive agents, yet each is composed of countless sand grains. Individual grains, when transported by sufficiently strong flows, self-organize into larger structures, many scales above the grain scale, to form bedforms such as ripples and dunes.

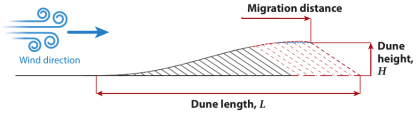
On earth, dunes develop naturally through aeolian, or wind-driven, processes (Andreotti et al., 2013), along riverbeds (Best, 2005), and in snow fields (Filhol and Sturm, 2015; Vriend and Bacik, 2025). The ubiquitous nature of dune formation and migration extends beyond Earth: to other planets (Bishop, 2007), moons (Diniega et al., 2017), and even a comet (Jia et al., 2017).

Bedforms range in scale, from 1 $cm$ subaqueous ripples, through 1 $m$ martian ripples (Alvarez et al., 2025), to 1 $km$ aeolian sand dunes. Countless characterizations have been given to dunes, including parabolic, transverse, barchan, dome, star, linear, and pancake (Courrech du Pont et al., 2024). These distinctions arise from variations in fluid and grain properties, sediment availability, vegetation, and the direction of forcing . Barchan dune fields are found in areas of unidirectional flow and limited sand availability (Bagnold, 1941). These crescent shaped dunes have a shallower stoss side upon which the fluid accelerates, carrying sediment (Charru et al., 2013). The crest marks the highest point of a dune, and is followed by the brink. Grains avalanche across the brink once the local accumulation surpasses the angle of repose–the critical angle at which granular material is stable Figure 1a: fluid streamlines respond to the steep change in topography face by separating, forming a recirculation bubble on the dune's lee side Figure 2. Downwind of the brink, the drag velocity decreases as the area is sheltered by the upwind dune (Bagnold, 1941). The flow reversal zone leaves an imprint on the sand, which can be interpreted through an inverse relationship between ripple wavelength and flow velocity (Charru et al., 2013). Separation decreases from the crest downwind along the slip side in field observations (Elbelrhiti et al., 2005) and wind tunnel experiments (Andreotti et al., 2002).

Dunes emerge from the transport of individual sediment grains, yet their motion and size are dictated by larger-scale processes. Their morophodynamics remain an active area of research. Previous works attribute dune formation to both the growth of a linear instability–which selects the most unstable wavelength–and to nonlinear processes, such as pattern coarsening (Andreotti et al., 2002; Vriend and Jarvis, 2018). Dune size is dictated by the saturation length, or the lag between changing flow conditions and the sediment response. Ripples,




**(a)** Key geometrical components




**(b)** Characteristic lengths and migration velocity direction.

**Figure 1:** Brachan dune geometry representations adapted from Vriend and Bacik (2025)

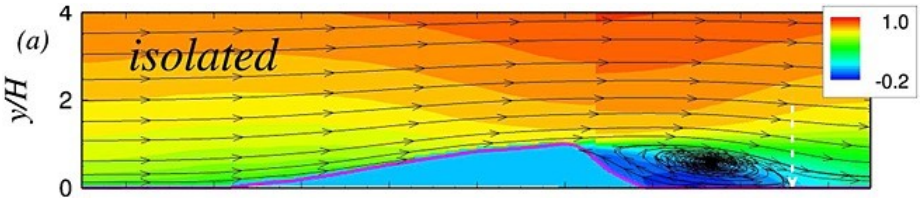



**Figure 2:** Adapted from Omidyeganeh et al. (2013) showing a large eddy simulation of the separation bubble over a single barchan.

smaller scale bedforms, more directly reflect the motion of individual grains. In a simplified sense, their wavelengths correspond to the characteristic length of a grain's saltation–or jump–which depends on the flow velocity (Bagnold, 1941).

**Why should we care to study dunes?** Dunes have a direct impact on human environment and livelihood; aeolian dune migration affects infrastructure Bruno et al. (2018), agriculture, and ecological systems. Changes in climate–particularly shifts in wind patterns–contributes to the expansion and activation of dune fields (Baas and Delobel, 2022). Climate conditions, such as drought, can reduce stabilizing vegetation levels, reactivating dormant dunes whose migration produces long-lasting alterations to regional soil composition (Ashkenazy et al., 2012; Lam et al., 2011). Increasing wind regimes are another climate-driven mobilizer of dormant dunes, further contributing to desertification (Thomas et al., 2005). A better understanding of the mechanisms behind dune migration will be increaingly critiacl as Earth's climate continues to change.

**The state of the art and a brief history of dune investigations** The classic text "*The Physics of Wind Blown Sand and Desert Dunes*" Bagnold (1941), is almost a century old, but remains an important reference work. Through extensive field work in North Africa and the Middle East, as well as wind tunnel experiments, R.A. Bagnold demonstrated that dune behavior could be predicted from fluid and grain mechanics and scaling laws. R.A. Bagnold

introduced one of the earliest sediment flux laws, which formed the basis for the first continuum saltation models. Other landscape-scale experiments have since been conducted, such as those of Hersen et al. (2002) at the Tarfaya Dune field in Western Sahara. Advances in satellite imagery now allow for the study of entire dune fields, as demonstrated by Durán et al. (2011) who observed dune field regulation through calving.

On a laboratory scale, aeolian dunes can be more difficult to reproduce compared to subaqueous dunes, since the latter have smaller temporal and spatial scales (Hersen et al., 2002; Hersen and Douady, 2005; MANTZ, 1978). By reproducing barchans in a water tank, Hersen et al. (2002) characterized differences in the morphological mechanisms between subaqueous and aeolian barchan dunes using the inertial length of sediment transport. Other experiments, such as the work of Endo et al. (2004), focused on multiple dune interactions. Previous work from our group used an annular flume to investigate the role of flow speed, bed thickness, and flow depth on dune evolution (Jarvis et al., 2022), as well as the influence of size ratio on dune collision types (Jarvis et al., 2023). With the same experimental system, Bacik et al. (2020) explored dune-dune repulsion, a non-contact mode of dune interaction.

In addition to physical systems, models have been developed as complementary options of exploring dune physics. Notable examples include: the numerical model of Hersen et al. (2004) based on their 2002 study Hersen et al. (2002), and a cellular automaton model developed by Katsuki et al. (2005), which was evaluated against Endo et al. (2004). The model of Katsuki et al. (2005) was later improved in Katsuki et al. (2011) to include saltation, strong erosion, and avalanching.

Accurate modeling of long term dune field behavior requires an understanding of dune-dune interactions. This work focuses specifically on interactions between barchan dunes, and examines how changing one of the main morphological control of dune field evolution–flow strength–affects the types of interactions between individual dunes. I identify two distinct flow regimes that yield a different set of dune interaction behaviors, and explore creative ways to model mid-interaction mechanics. Prior to beginning my two-dune investigation, I verify my techniques using a one dune setup.

This current work utilizes the open-source cellular automaton model, ReSCAL, to further understand processes driving dune migration and interaction. The thesis is structured as follows: Section (1) includes the introduction and methods overview. Section (2) explores one dune simulations, Section (3) examines a two-dune system, followed by a concluding Section (4).

## 1.2 ReSCAL: a Real-Space Cellullar Automaton Laboratory

Dune migration has been modeled through a variety of continuous and discrete methods, each posing a different set of strengths. The general challenge faced by most models arises in relating small-scale interactions to large-scale patterns (Rozier and Narteau, 2014a).

**Continuum saltation** models connect the sand flux to fluid velocity through partial differential equations. The simplest flux law, which gives a cubic relation between shear velocity and sand flux, was introduced by R.A. Bagnold in 1936. Since then, complexity of such laws has increased, and the corresponding partial differential equations have been integrated into more current continuum saltation models such as Sauermann et al. (2001). These models frequently make simplifications regarding the uniformity of the flow, and do not reflect grain size interactions or processes.
**Discrete Element Models**, such as presented in Durán et al. (2014) resolve individual grain movements and give insight into sediment transport mechanics. These models quantify dynamical interactions between topography, shear stress, and sediment transport, but are computationally expensive and often simplify the flow over a complex topography (Narteau et al., 2009).

On the other hand, **CFD models** represent the turbulent nature of the flow well(Omidyeganeh et al., 2013), and have been used to simulate barchan collisions (Zhou et al., 2019). However, computational cost of such models does not make them advantageous for examining long-term dune field evolution patterns.

**Cellular automaton (CA)** models are discrete models, where the space is divided into a 2-D rectangular lattices, and sediment height is measured according to the accumulation of "sand slabs" (Narteau et al., 2009). Transition rules dictate how the slabs evolve throughout spatially throughout the grid, and temporally across timesteps. Relating simulation outcomes to physical process has been a persistent issue in cellular automaton modeling Chopard and Droz (1998). Rate parameters mimic various sediment movements, such as erosion ($\lambda_e$) and deposition ($\lambda_c$). However, their actual values have no connection to real sediment motion on the grain scale. Another drawback of many past cellular automaton models, such as the one presented in Katsuki et al. (2011), is the inability to model unsteady flow conditions.

**Real-Space Cellular Automaton Laboratory (ReSCAL)** solves the issue of connecting model length and time scales to physical ones. The sediment model is coupled with a Lattice Gas Cellular Automaton (LGCA) model to incorporate the effect of turbulent flow on sediment transport. Full model specifics are outlined in Narteau et al. (2009); Rozier and Narteau (2014a, 2012), but a

summary is included in the section below.

### 1.2.1 The Sediment Model:

The model space is divided into cells of length $l_0$. Cells are in one of the following states: fluid, mobile sediment, immobile sediment, or neutral. Neutral cells represent the domain boundaries (Jarvis et al., 2023). Physical processes of sediment transport (erosion, deposition, transport) are modeled through nearest neighbor interactions of doublets (Figure 4). All transitions acting on a pair of cells, or a doublet, are assigned a specific transition rate for each orientation. Each process has an associated characteristic timescale, expressed in units of $t_0$ (the model time step). Every transition has a certain probability of occurring, making the model not entirely deterministic, and requires a random seed as an input.

Figure 3 outlines the main iteration scheme, which is also described below. The algorithm is utilizes a Makov Chain, in which a single transition is associated with a single active doublet. The information originates from Rozier and Narteau (2012).

**Initialization:**

Initialization of Transitions:
Transitions correspond to physical processes of sediment transport, such as erosion and deposition.

In ReSCAL, a transition contains the following information:

- the initial state of the doublet: $(S_1^i, S_2^i)$, with each state $(S)$ corresponding to a cell.
- the final state of the doublet: $(S_1^i, S_2^i)$
- orientation: $(+x,\ +y,\ +z)$
- the transition rate, $\lambda$, which determines the probability of the specific transition occurring.

Initialization of Cellular Space:
The initial configuration of the cellular space is created. Each cell is one of the following:

- $0 \rightarrow$ immobile sediment

- $1 \rightarrow$ mobile sediment
- $2 \rightarrow$ fluid
- $3 \rightarrow$ neutral (for boundaries)

In my simulations, the dunes are initialized as triangular heaps of sediment.

Initialization of Generic Doublets:
A *Generic Doublet* behaves as a template for any possible doublet configurations, and contains the state of each cell $(S_1, S_2)$, as well as their orientation.

An *Active Generic Doublet (AGD)* is a generic doublet that matches an initial configuration $(S_1^i, S_2^i)$specified by at least one transition.

The cellular space is searched, and any *Active Doublets (ADs)* that match an AGD are referenced by the doublet array of that AGD. The doublet array contains the position of the first cell $(S_1)$ in the AD, as well as a pointer to the cell array. Only the position $(S_1)$ is needed since the AGD contains an orientation. The cell array contains a maximum of three elements, one for each possible orientation. Each element holds the cell's index in the doublet array, and a pointer.

**Iteration:**
Each timestep, each of the following steps occur.

Weighted Random Choice of Transitions
A transition is chosen based on set rates $(\Lambda_i' s)$, and is described by (1). Resultingly, an AGD corresponds to the chosen transition.

$$w_i(t) = \frac{N_i(t)\Lambda_i}{\Lambda(t)} \tag{1}$$

where $N_i(t)$: number of active doublets for the transition $T_i$.

The algorithm can be characterized as a generalized Poisson process or a type of continuous time Markovian process. Resultingly, the **time between transitions** is described by (2).

$$\Lambda(t) = \sum_{i=1}^{n} N_i(t)\Lambda_i \tag{2}$$

Random Choice of Active Doublet
From the doublet array of the Active Generic Doublet, an Active Doublet in the cellular space is randomly chosen.

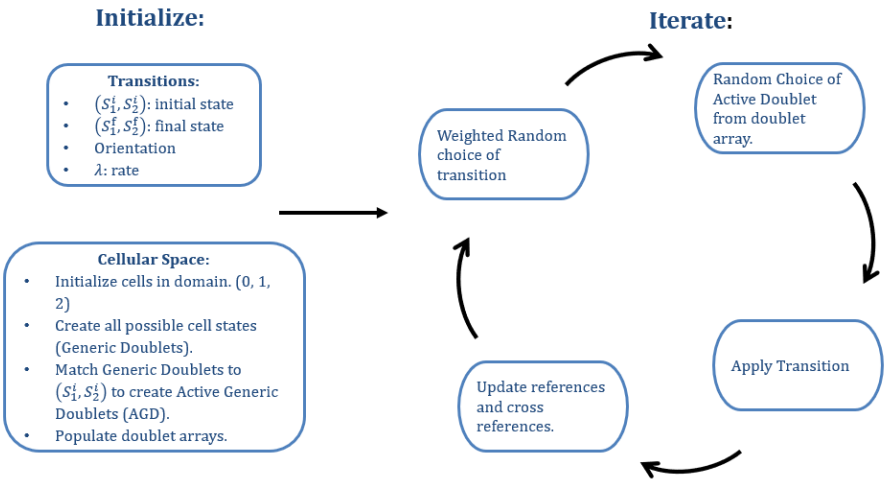



**Figure 3:** Algorithm of the sediment model

Applying transition
The transition is applied and the states $(S_1, S_2)$ of the active doublet may change.

Updating References
If changed, the AD is removed from the doublet array of its old AGD. Neighboring cells are inspected, the doublet arrays of any newly relevant AGDs are updated, and the corresponding cell cross-references are revised.

The Passage of Time
The **waiting time** before the next transition is a random variable set by (3), where p is a random variable between 0 and 1.

$$\Delta t = -\frac{1}{\Lambda(t)} \ln(1-p) \tag{3}$$

### 1.2.2 How is the flow simulated? : The (LGCA) model

A lattice gas cellular automaton (LGCA) model, described in Rothman and Zaleski (2004), solves the Navier Stokes equation in 2D Binnig et al. (1986). The model runs simultaneously with the evolution of the topography, and allows for the determination of the bed shear stress. The bed shear stress, $\tau$ connects to the sediment flux reported in the sediment model through (7) (Narteau et al., 2009; Rozier and Narteau, 2014b).

**Model specifics:** The space is divided into a set of uniformly spaced vertical planes parallel to the direction of the flow. Inside the plane, "Boolean molecules" with associated velocity vectors move from cell to cell.

A single iteration of the LGCA model comprises of a propagation phase followed by a collision phase. Initially, particles remain in place or move to their neighboring or nearest neighboring cell depending on their velocity vectors, with magnitudes of : 0, 1, $\sqrt{2}$. During the collision phase, particles on the same cell may exchange momentum - and as a result their velocity vectors - according to imposed collision rules described in Rothman and Zaleski (2004). Conservation of mass (number of particles) and momentum (sum of velocity vectors) allow for the 2D LGCA model to simulate flow based on the 2D Navier Stokes equation.

The flow is calculated at a much shorter timescale than what is required to move a sedimentary cell, which allows for the simultaneous coupling of the two models. Particles are averaged over cells and times steps, and an averaged velocity field, expressed as a number of fluid particles, is used to calculate the bed shear stress, $\tau_s$, through (4). $\tau_0$ is the stress scale of the model.

$$\tau_s = \tau_0 \frac{\partial \vec{V}}{\partial \vec{n}} \tag{4}$$

Although the shear stress, $\tau_s$, is a model output, the critical shear stress described in (7) is used to control the flow speed. A higher critical shear stress, $\tau_c$ or $\tau_1$, corresponds to less sediment flux, and an effective lower flow rate. The highest simulated flow velocity in the model is achieved by setting $\tau_1$ to 0. The erosion rate, described by (5) , is another example of a coupling between the flow and the bed topography.

$$\Lambda_e = \begin{cases} 0 & \text{for } \tau_s \leq \tau_1 \\ \Lambda_0 \frac{\tau_s - \tau_1}{\tau_2 - \tau_1} & \text{for } \tau_1 \leq \tau_s \leq \tau_2 \\ \Lambda_0 & \text{else,} \end{cases} \tag{5}$$

The rebound dynamics of the velocity particles depend the on the sedimentary cells and are another place of coupling between the two models. The normal to the topography, $\vec{n}$, is perpendicular to surface and calculated using differences in heights of cells located $2l_0$ in all directions.

**Boundary conditions:** The LGCA model utilizes a free slip boundary condition along the top of the domain: depending on the angle $\theta$ between colliding particle and the normal to the ceiling is between $-\pi/8 \quad \pi/8$, particle return along the path from which they approach. Otherwise, the particle rebounds with a rotation of $\pm\,\pi/2$ in a perpendicular direction according to the sign of $\theta$. Along the bottom, a turbulent boundary layer is represented through no slip boundary condition with a small scale of roughness, ensuring the direction of the normal changes along the x direction. Periodic boundary conditions along the

horizontal sides of the domain cause particles leaving the domain to reenter on the opposite side. Lastly, a permanent forcing simulated unpertubated flow far above the bed. With about a 0.2 % chance each iteration, fluid particles moving in the opposite direction to the flow are randomly replaced by ones moving in the correct direction.

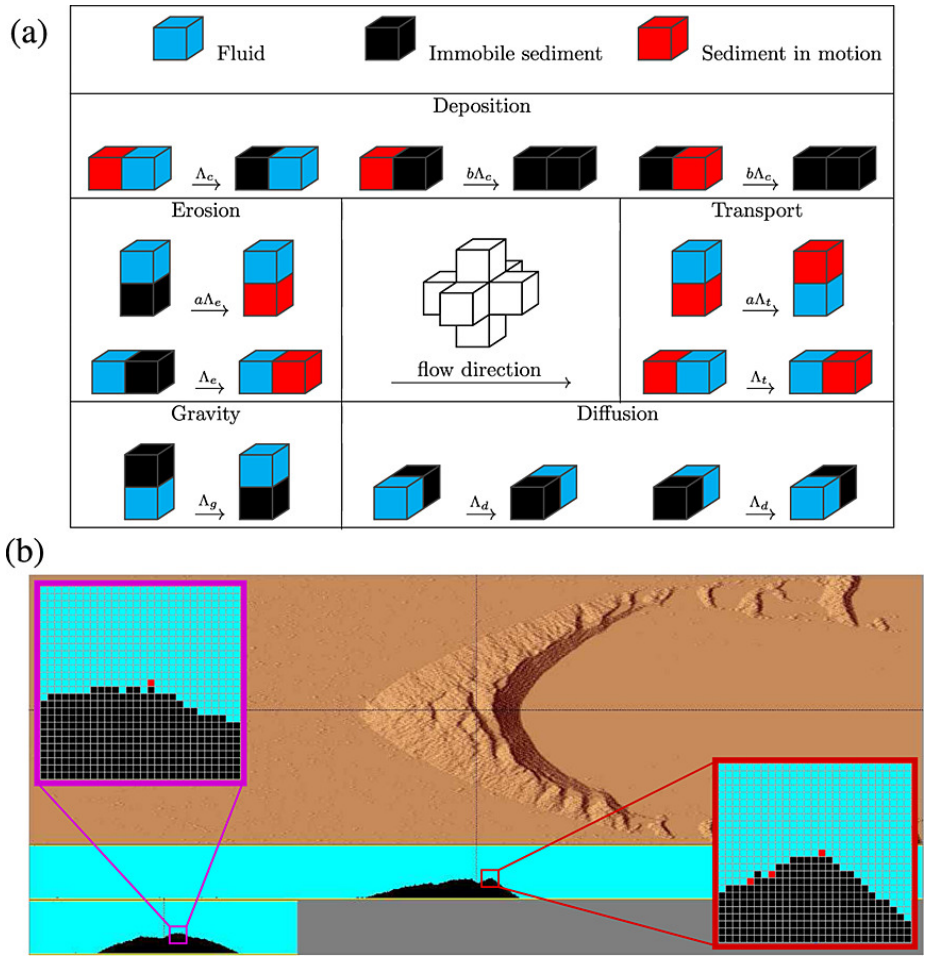



**Figure 4:** Adapted from Narteau et al. (2009): A CA dune model using ReSCAL. (a) In the CA model, three states are used to reproduce the fluid, mobile and immobile sediment. All transitions acting on a doublet are given a specific rate parameter ($\lambda$). (b) Topography of a barchan dune in the CA dune model. Dashed line corresponds to the longitudinal and the transverse vertical slices of cells shown below.

**On avalanches:** To incorporate the physical process of avalanching - which

occurs when enough sediment builds up along the brink, eventually exceeding the angle of repose - ReSCAL includes an avalanche rate parameter: $\Lambda_{ava}$. The actual transition rate of avalaching, $\lambda_\theta$, is calculated using (6).

$$\Lambda_\theta = \Lambda_{ava}\delta_\theta \quad with \quad \delta_\theta = \begin{cases} 0 & \text{if } \theta \leq \theta_c \\ 1 & \text{if } \theta > \theta_c \end{cases} \tag{6}$$

## 1.3 Input and Output Parameters

An example set of parameters is given in Table 1. I conducted sensitivity tests on the rate parameters ($\lambda$'s), but results did not significantly vary from the set shown in Table 1.

## 1.4 Converting Model Values to Physical Values

The outputs of the numerical simulation can be compared to real bedforms, allowing the length and time scales of the models can be computed through a linear stability analysis (Rozier and Narteau, 2014b). Beginning from a flat sand bed, the model generates periodic dune patterns. The most unstable mode reached for a wavelength $\lambda_{max}$ scales with the ratio of sediment and fluid densities, times the grain diameter (Hersen et al., 2002).

**Setting the Length and Time Scale** The flat bed experiment allows for the setting of the length ($l_0$) and time ($t_0$) scales of the model for a certain set of parameters. Model values can be converted to physical units using the calibrated $l_0$ and $t_0$. Figure 5b, Figure 7 in Narteau et al. (2009), displays the results of one such experiment. The following section describes the process of obtaining the length and time scales of my own simulations. The process is based on the one outlined in Zhang et al. (2014). Flow of varied strength was simulated across a flat bed lacking sediment. Outputted surface shear stress data was converted to saturated flux through Equation (7).

$$Q_{sat} = \begin{cases} 0 & \text{if } \tau_s \leq \tau_c, \\ \tau_s^\gamma(\tau_s - \tau_c) & \text{if } \tau_s \geq \tau_1. \end{cases} \tag{7}$$

where $\gamma$ is a constant equal to 1 in the case of aeolian sand dunes (Bagnold, 1941). The minimum surface shear stress, $\tau_1$ , is a user-set parameter in the ReSCAL, and $\tau_s$ is the outputted surface shear stress.

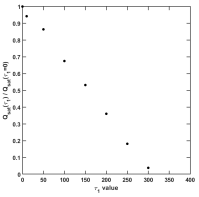
Rearranging Equation (8), we can calculate the corresponding flow velocity, $u_*$

| Symbol | Description | Model Value | Physical Value |
|---|---|---|---|
| $l_0$ | length scale (cell size) | - | $0.5124\,m$ |
| $t_0$ | timescale | - | $14884.86\,s$ |
| $H$ | domain height | $300\,l_0$ | $130.2\,m$ |
| $L$ | domain length | $2000\,l_0$ | $868\,m$ |
| $\Lambda_e$ | transition rate for erosion | $4\,t_0^{-1}$ | varies depending on $\tau_1$ |
| $\Lambda_c$ | transition rate for depostition | $2\,t_0^{-1}$ | varies depending on $\tau_1$ |
| $\Lambda_t$ | transition rate for transport | $6\,t_0^{-1}$ | varies depending on $\tau_1$ |
| $\Lambda_d$ | transition rate for diffusion | $0.02\,t_0^{-1}$ | varies depending on $\tau_1$ |
| $\Lambda_g$ | transition rate for gravitational settling | $10\,t_0^{-1}$ | varies depending on $\tau_1$ |
| $\Lambda_a$ | transition rate for avalanching | $10\,t_0^{-1}$ | varies depending on $\tau_1$ |
| $\theta$ | angle of repose | $35°$ | $35°$ |
| $\tau_1$ | critical shear stress for motion - adjusted to vary flow condtions. | $0\,\tau_0 - 300\,\tau_0$ | - |
| $\tau_2$ | slope parameter (maximum shear stress value). | $1000\,\tau_0$ | - |

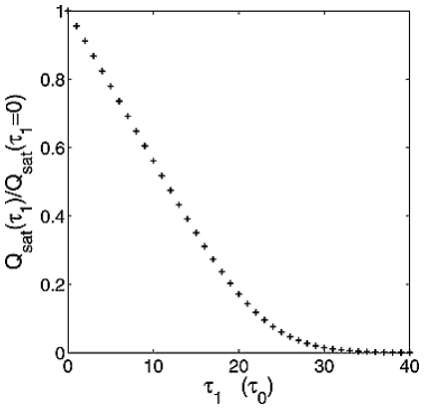
**Table 1:** An example set of parameter values. The above were used Section 4, as well as the migration velocity analysis. Values were adapted from Jarvis et al. (2023), and procedure for physical conversions is outlined in the Flat Bed Section.

**Table 2:** Values of fluid and grain properties, from literature and theory.

| Variable | Description | Value | Units |
|---|---|---|---|
| $\rho_f$ | air density on Earth | 1.293 | $\frac{kg}{m^3}$ |
| $\rho_{f,w}$ | water density on earth | $10^3$ | $\frac{kg}{m^3}$ |
| $\rho_s$ | sand density on Earth | $2.55 \cdot 10^3$ | $\frac{kg}{m^3}$ |
| d | sand diameter on Earth | 200 | $\mu m$ |
| g | gravity on Earth | 9.81 | $\frac{m}{s^2}$ |
| $\sqrt{\Theta_t}$ | square root of the Shields number | 0.082 | |
| $\alpha_q$ | constant | 8.33 in air ( 2 for water) | |



**(a)** Flatbed simulation results under my simulation parameters.




**(b)** Figure 7, Narteau et al. (2009). $\tau_1$ values are scaled by a factor of 10.

**Figure 5:** The saturated flux with respect to the $\tau_1$ value. $Q_{sat}(\tau_1)$ is normalized by its value $Q_{sat}(\tau_1 = 0)$. Note that an increasing $\tau_1$ value corresponds to a decreasing flow strength.

through Equation (9).

$$\frac{Q_{sat}(\tau_1)}{Q_{sat}(\tau_1 = 0)} = 1 - \left(\frac{u_c}{u_*}\right)^2 \tag{8}$$

$$u_* = \frac{u_c}{\sqrt{1 - \frac{Q_{sat}(\tau_1)}{Q_{sat}(\tau_1=0)}}} \tag{9}$$

The threshold velocity, $u_c$, is calculated in (10). Values of the physical constants are shown in 2. The physical saturation flux, for a given ratio of $\frac{Q_{sat}(\tau_1)}{Q_{sat}(\tau_1=0)}$ , can be calculated using (11).

$$u_c = \sqrt{\Theta_t}\sqrt{\frac{\rho_s - \rho_f}{\rho_f} g d} \tag{10}$$

$$Q_{sat,p} = \alpha_q \frac{\rho_f}{\rho_s d} u_c \left(u_*^2 - u_c^2\right) \tag{11}$$

The time scale is calculated using Equation(12). To obtain one time scale value for the model, a reference ratio of $\frac{Q_{sat}(\tau_1)}{Q_{sat}(\tau_1=0)}$ was chosen (0.95), based on the value used in Zhang et al. (2014).

$$t_0 = \frac{Q_{sat}(model)}{Q_{sat,p}} l_0^2 \tag{12}$$

The length scale is calculated based on the linear instability analysis outlined in Narteau et al. (2009). The most unstable wavelength, $\lambda_{max}$, is the fastest growing wavelength of bed perturbation, and sets dune spacing. Equation (13) describes $\lambda_{max}$ as a function of the physical parameters, and the length scale, $l_0$ is found through Equation (14) .

$$\lambda_{max} = 50d\frac{\rho_s}{\rho_f} \tag{13}$$

$$l_0 = \frac{\lambda_{max}}{40} \tag{14}$$

Depending on the system to which ReSCAL is compared, the length and time scales will change. The length scale for my aeolian simulations had a value of $0.5124m$ and the time scale was approximately 4 hours ($14884.8s$). These values are of the same magnitude as reported in Zhang et al. (2014). When the code was modified for aqueous dune settings, the resulting time and length scales were: $0.0007\,m$ and $5.86s$ .

Figure 5a captures the recreation of the flatbed experiment that Narteau et al. (2009) utilizes to find the length and time scale of the ReSCAL Figure 5b. The similarity between the two figures motivates any conversions to physical values that I would conduct in my implementations of ReSCAL.

**Disclaimer** : Unfortunately, my research was purely simulation based and did not have a physical component to compare to. However, recreating the flat bed experiment described in the section is motivation for future work to examine my experiments using physical data as well.

### 1.4.1 Saturation Length

In Section 3, the location of the maximum sediment flux, $Q_{sat}$, on a steady state dune is examined under varying conditions. The plots shown in Figure 8, Figure 9 show the location of $Q_{sat}$ to be both upwind and downwind of the surface shear stress maximum, $\tau_{s,max}$. While both maximums display oscillating behavior, confirmed by Figure 10, one would expect the maximum sediment flux to at least be in line with the maximum forcing. The saturation length, $l_{sat}$, resolves the unphysical appearance of $Q_{sat}$ upwind of $\tau_{s,max}$ in a given timestep.

The shear stress maximum is generally located upstream of the crest (Vriend and Jarvis, 2018). Consequently, due to fluid inertia, the flow accelerates upwind of the crest and decelerates on the downwind slope (Vriend and Jarvis, 2018). Sand is eroded upwind of the crest and deposited on the downwind lee slope. The maximum sediment flux, or saturated flux, is downstream of the maximum surface shear stress. The characteristic length scale in a dune: the

saturation length ($l_{sat}$), represents the lag in the response of the sediment flux to a local change in surface shear stress (Bagnold, 1941; Elbelrhiti et al., 2005; Narteau et al., 2009). In ReSCAL, the model length scale, $l_0$, is related to the saturation length: $l_{sat} \approx 6.9 l_0$, through a linear stability analysis on a flat bed. The saturation length of our aeolian model - $l_0 = 0.5124\, m$ - agrees with the $l_0$ value reported in Narteau et al. (2009), which suggests our model saturation length is $l_{sat} \approx 6.9\, l_0$, depending on the flow strength.

# 2 One Dune Morphing From triangular Heap

## 2.1 Effect of Varied Flow Speed on Dune Shape and Speed

Under a constant dune area of 6000 $l_0^2$, the minimum surface shear stress was changed to simulate varied flow speeds. A lower $\tau_1$ value corresponds with a higher flow velocity. In each case, steady state was achieved.

Six critical shear stress values values ($\tau_1 = 1$, $\tau_1 = 50$, $\tau_1 = 100$, $\tau_1 = 150$, $\tau_1 = 200$, $\tau_1 = 300$) are simulated. Due to the probabilistic nature of ReSCAL, each case consists of fifty individual simulations with a random seed. The dune was auto-centered every 50 $t_0$, with data collection every 100 $t_0$. The auto-centering allows for averaging across time steps. Ten simulations in each $\tau_1$ set were run for 50,000 $t_0$ to ensure steady state migration velocity was reached. The remaining forty ran for 10,000 $t_0$.

The aim of this section of the project was to investigate whether the relative locations of the surface shear stress and sediment flux change under different flow speeds, or if only their respective magnitudes differ.

### 2.1.1 Effect on topography

The aspect ratio of dune height to dune length characterizes the effect of varied flow speeds on dune topography. Visual inspection of Figure 8 and Figure 9, and aspect ratio values shown in 3 support the "stretching" of a dune under lower flow conditions.

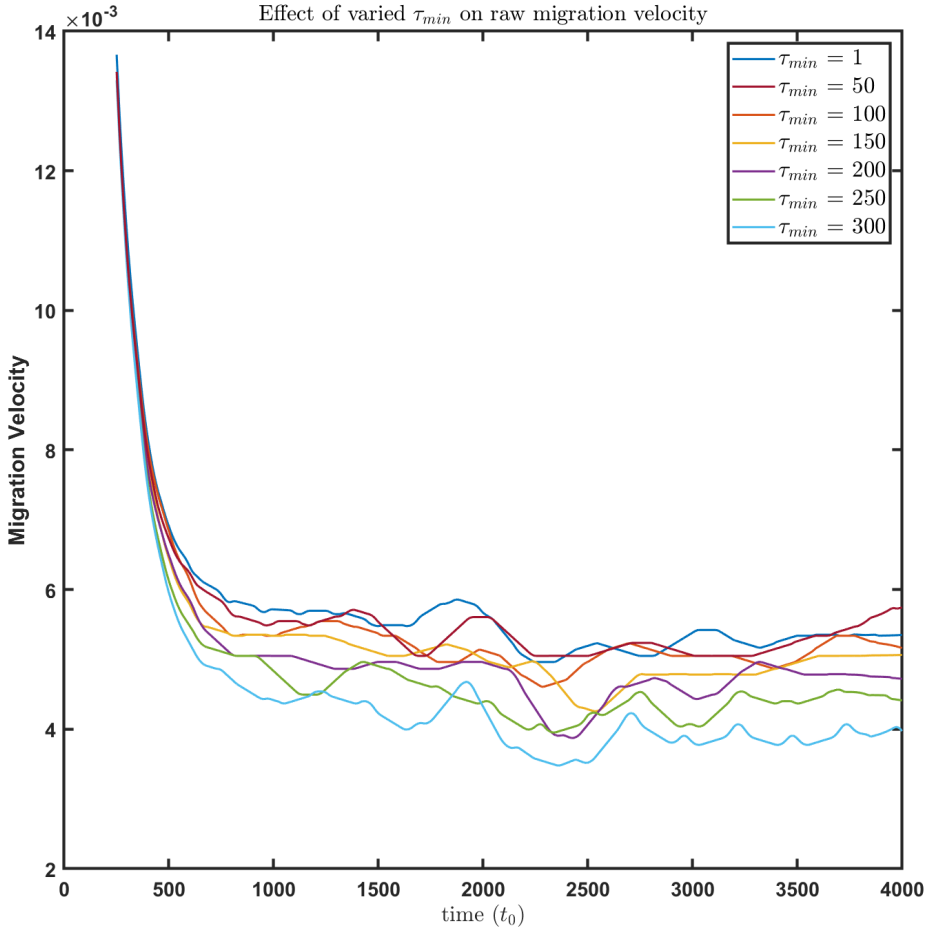



**Figure 6:** Migration velocities of dunes with cross sectional area 6000 $l_0^2$ under varied flow conditions.

### 2.1.2 Effect on migration velocity

Figure 6 shows the migration velocities over time of six different triangular heaps of the same area (6,000 $l_0$), as they morph into a dune shape. Each case represents a different critical shear stress ($\tau_1$). Data from fifty simulation results was averaged and smoothed over space for each flow velocity (or $\tau_1$) condition. To find the dune migration velocity, the dune toe (or downwind edge) is tracked throughout time. A clear trend of decreasing migration velocity with decreasing flow strength (increasing $\tau_1$) can be noted.

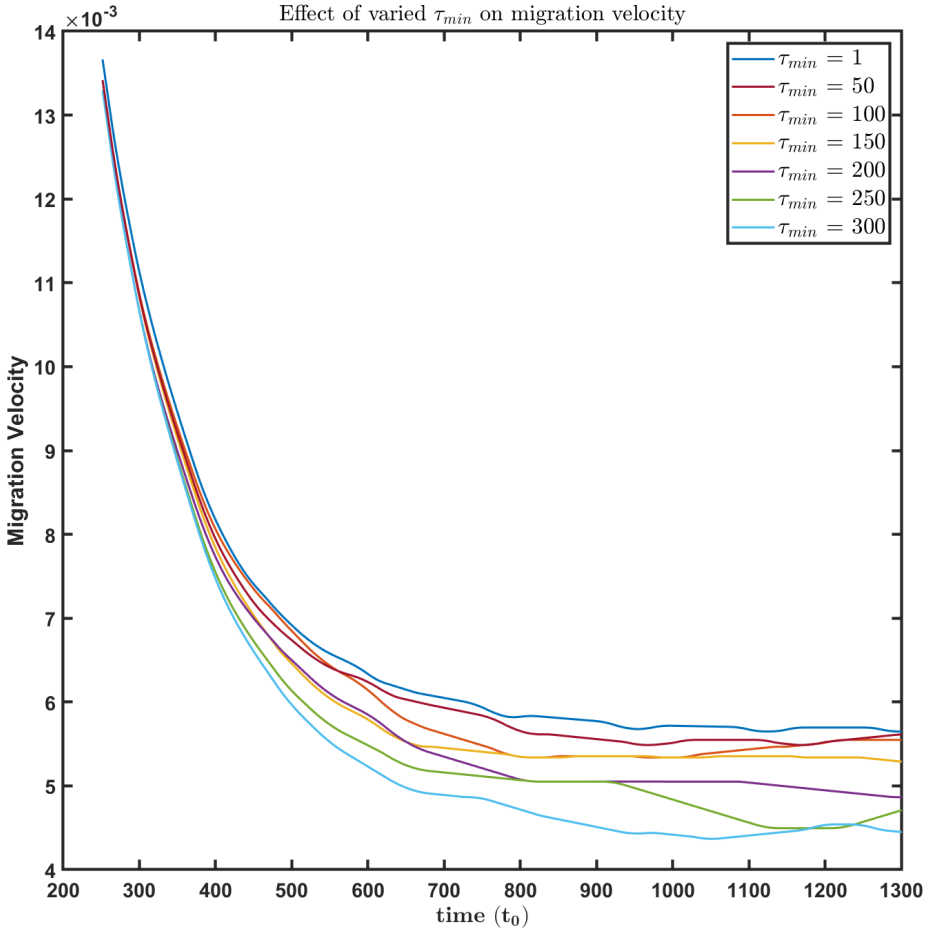



**Figure 7:** The transition to steady state of the dunes with cross sectional area 6000 $l_0^2$ under varied flow conditions.

### 2.1.3 Time to reach steady state

Figure 7 displays the transition to steady state of seven $\tau_1$ cases. All cases show a sharp decrease in migration velocity, followed by a slower decrease, until steady state is reached. The lines follow an exponential decay path and rarely cross. The few crossing are attributed to noise. The slowest flow velocity case experiences the steepest initial decline as it undergoes the most physical change in shape Figure 9. The largest flow velocity experiment remains above the other lines in the dataset. If the limited crossing of lines can be attributed to noise, a claim can be made that since all seven cases eventually reach a semi-horizontal migration rate, the slowest flow velocity case takes the longest to reach the steady state value, while the fastest flow velocity case takes the least amount of

time.

### 2.1.4 Effect on Location of Maximums

Figure 6 depicts the migration velocities of the dunes and confirms the steady state assertion. The plots shown in Figure 8 and Figure 9 represent averages over fifty simulation runs. For each figure, the positions of the maximum topography (crest), maximum surface shear stress ($\tau_{s,max}$), and the maximum sediment flux ($Q_{max}$) are shown.

It is important to note that $Q_{max}$ does not correspond to the location of the saturated flux. In ReSCAL, the instantaneous flux is measured as the number of cells moving across a specified spatial location. The saturation length $l_{sat}$ is inherently specified in the model and is most clearly observed through the average lag between the locations of maximum shear stress and maximum sediment flux. This lag is examined further in Section 3.2.5.

For each flow strength, Table 3 reports the relative distance from the brink to each maximum at the same timestep shown in Figure 8 and Figure 9. To form a more representative assessment of the effect of flow strength on the relative locations of these maxima, the $x$-coordinates were averaged over time, as shown in Table 4. These averaged values were determined using a standard error analysis for each case of $\tau_1$, with Figure 10 shown as an example.

The time-averaged results indicate a systematic shift in the relative positions of the maxima with changing flow strength. Under high flow conditions, the maximum shear stress, $\tau_{s,max}$, aligns closely with the crest and remains only slightly upwind of it. As flow velocity decreases, $\tau_{s,max}$ shifts progressively farther upwind. Likewise, the distance between $\tau_{s,max}$ and the maximum sediment flux, $Q_{max}$, increases as flow strength decreases.

As shown in Figure 8, Figure 9, lower flow velocities are associated with a shallower stoss face. This flattening is consistent with a reduction in grain accumulation near the crest, which may alter the nature and frequency of avalanching on the lee side. Under slower flow conditions, avalanches are expected to originate closer to the brinkNield et al. (2017). On the other hand, stronger flow conditions promote multiple failures. Under high flow velocities of 8 $m/s$, Nield et al. (2017) observed primary grainfalls originating near the top of the slip face, and secondary grainfalls occurring as far out as 6.4 m downwind. Previous studies have also identified a correlation between wind velocity and avalanche frequency, although the relationship between wind velocity and avalanche magnitude remains unresolvedSutton et al. (2013b,a).

Table 3 further shows that the crest shifts further upwind as flow strength decreases. This trend is consistent with the increased crest-brink spacing observed

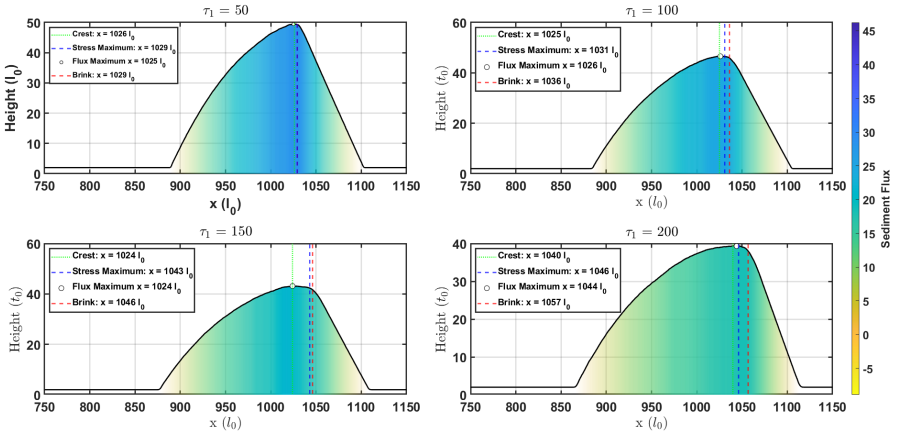



**Figure 8:** Dunes of cross sectional area 6000 $l_0^2$ under various flow conditions achieved by increasing the $\tau_1$ parameter. The color gradients shows the level of sediment flux at that location. The x-coordinates of the maximum topography (crest), maximum surface shear stress, and the saturation flux are plotted.

in lower flow cases, and with the overall reduction in aspect ratio shown in in Figure 8 and Figure 9. A larger crest-brink separation increases the area over which sediment can accumulate before avalanching occurs. Because avalanching is initiated once local slopes exceed the angle of repose, a broader accumulation zone may spread deposited grains over a larger region and thereby reduce the rate of lee-side grainfall. However, these results do not by themselves allow for a definitive conclusion regarding avalanche magnitude under varied flow conditions.

The observed behavior also has implications for dune migration. In order for the dune to advance, sediment transported over the crest must reach the downwind toe through grainfall down the slip face. This interpretation is consistent with the migration metric used in this work, which tracks the position of the downwind dune toe.

### 2.1.5 Standard Error Analysis

An analysis of the standard error, SE, is displayed in Figure 10. The standard error is defined by Equation (15): the ratio of standard deviation, $\sigma$, to the square of the sample size, $n$. Figure 10 examines the error in locations of maxima for $\tau_1 = 100$ and $\tau_1 = 200$. A clear lag can be observed between the maximum sediment flux, $Q_{max}$, and the maximum shear stress, $\tau_{s,max}$. The

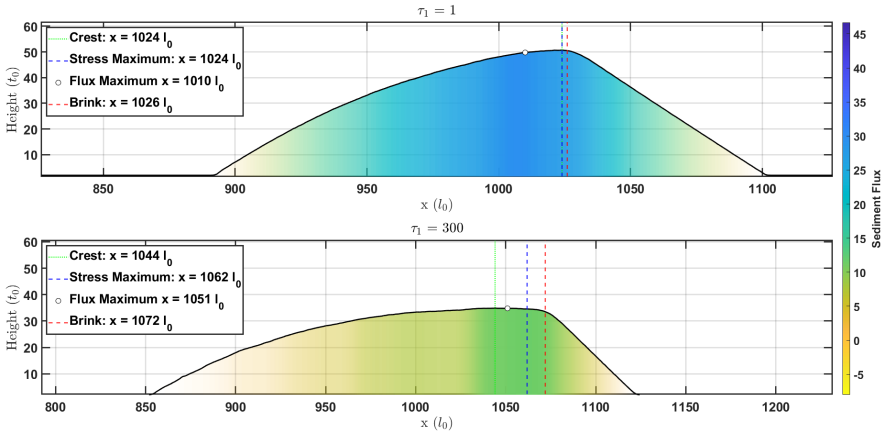



**Figure 9:** Dunes of cross sectional area 6000 $l_0^2$ under the more extreme flow conditions: $\tau_1 = 1$ (fast flow) and $\tau_1 = 300$ (slow flow).

| **Flow Strength ($\tau_1$)** | **Distance of $Q_{max}$ from brink** | **Distance of$\tau_{s,\max}$ from brink** | **Distance of crest from brink** | **Aspect Ratio ($\frac{\mathbf{H}}{\mathbf{L}}$)** |
|---|---|---|---|---|
| 1 | $0\,l_0$ | $-3\,l_0$ | $-6\,l_0$ | 0.2377 |
| 50 | $-3\,l_0$ | $-6\,l_0$ | $-9\,l_0$ | 0.2246 |
| 100 | $-4\,l_0$ | $-7\,l_0$ | $-9\,l_0$ | 0.1954 |
| 150 | $-25\,l_0$ | $-9\,l_0$ | $-18\,l_0$ | 0.1694 |
| 200 | $-33\,l_0$ | $-16\,l_0$ | $-30\,l_0$ | 0.1363 |
| 300 | $-57\,l_0$ | $-19\,l_0$ | $-62\,l_0$ | 0.0854 |

**Table 3:** Relative distance of maxima shown in Figures (8) and (9) relative to the dune brink. it is important to note that these referenced figures, as well as the reported maxima coordinates are taken from a single timestep.

| **Flow Strength** ($\tau_1$) | **Mean** $Q_{max}$ **Location** | **Mean** $\tau_{s,\max}$ **Location** | **Mean Crest Location** | **Approx.** $L_{sat}$ |
|---|---|---|---|---|
| 1 | $1018.69\,l_0$ | $1022.47\,l_0$ | $1022.86\,l_0$ | $3.78\,l_0$ |
| 50 | $1021.90\,l_0$ | $1025.46\,l_0$ | $1026.27\,l_0$ | $3.56\,l_0$ |
| 100 | $1026.18\,l_0$ | $1029.53\,l_0$ | $1031.82\,l_0$ | $3.35\,l_0$ |
| 150 | $1031.96\,l_0$ | $1036.93\,l_0$ | $1041.14\,l_0$ | $4.97\,l_0$ |
| 200 | $1040.75\,l_0$ | $1049.28\,l_0$ | $1056.61\,l_0$ | $8.53\,l_0$ |
| 300 | $1057.53\,l_0$ | $1083.20\,l_0$ | $1096.97\,l_0$ | $25.67\,l_0$ |

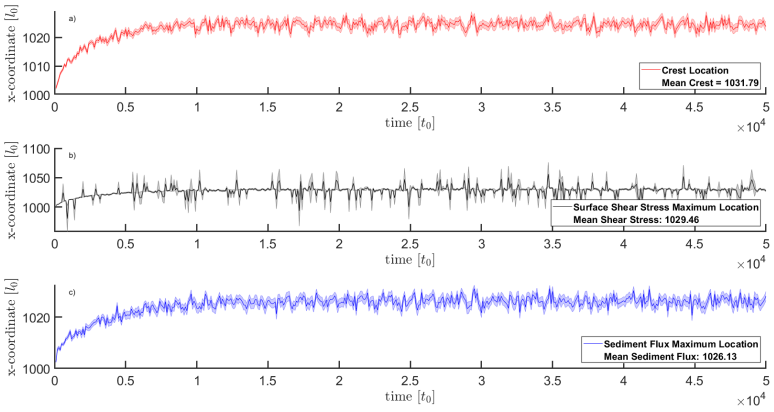
**Table 4:** Mean maxima locations from the error analysis of each $\tau_1$ case.

saturation length, $l_{sat}$, described in Section 1.4.1 can be observed in this lag.

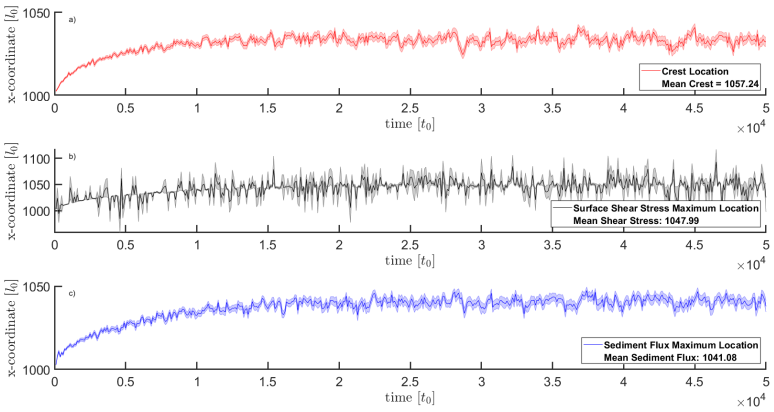
$$SE = \frac{\sigma}{\sqrt{50}} \tag{15}$$

**Implication for examining maximums at individual time-steps** Earlier in the section, Figure 8 and Figure 9 examined the locations of various maxima at a single timestep. Due to the oscillatory behavior of the reported mean, maxima identified at individual timesteps may not be representative of the systems as a whole, even when the selected timestep falls within the steady state regime.

**Another possible classification of steady state** The standard error analysis for six cases of $\tau_1$ is presented in Figure 10, Figure 29, and Figure 30. As $\tau_1$ increases, indicating a reduction in flow velocity, the spatial positions of the crest, maximum sediment flux, and maximum shear stress take progressively longer to reach steady values. According to the geometric steady state, the case with the fastest flow ($\tau_1 = 1$) reached a steady state at approximately $10,000\,t_0$, whereas the case with the slowest flow ($\tau_1 = 300$) necessitates approximately $400,000\,t_0$. In contrast, when steady state is assessed using migration velocity, dunes attain a steady migration rate before $2000\,t_0$ Figure 6, consistent with previous studies such as Jarvis et al. (2023). The findings exemplify different facets of dune evolution, each characterized by its respective steady state. They demonstrate the inherent method-dependency of the time required to reach the steady

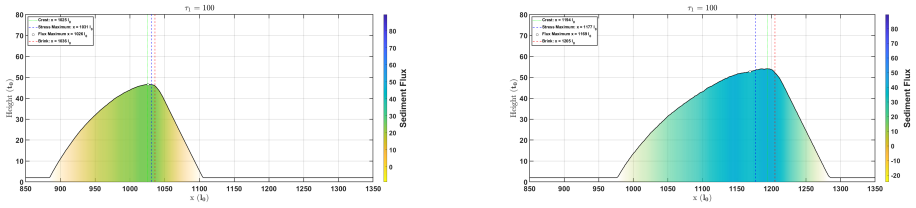



**(a)** $\tau_1 = 100$, with $l_{sat} \approx 3.35\, l_0$




**(b)** $\tau_1 = 200$ with $l_{sat} \approx 8.53\, l_0$

**Figure 10:** Standard Error across fifty simulations. Mean values are calculated from timestep 25,000 onward.




**(a)** Area 6000 $l_0$ **(b)** Area 10000 $l_0$

**Figure 11:** Comparison of varied area dunes (6000 $l_0$ and 10000 $l_0$ under the same flow conditions of $\tau_1 = 100$. Both have achieved steady state. Dunes are colored by magnitude of sediment flux, and the x locations of various maximums are shown in each plot.

### 2.1.6 Conclusion

The dune behavior under lower flow conditions is consistent with expectations, exhibiting both slower migration and a reduced aspect ratio, as shown in Figure 8 and Figure 9. This agreement with expected behavior supports the use $\tau_1$ as a meaningful control parameter in the remainder of the work.

The error analysis showed that a single timestep analysis of maximum locations may not adequately represent the system, since the locations of the various maxima oscillate over time. Additionally, two distinct definitions of steady state were identified, each reflecting a different method of characterizing dune evolution. This suggests that dunes may attain a constant migration constant rate even as their geometry continues to evolve.

## 2.2 Effect of Varied Dune Area

Another goal of the one dune study was to check the scaling invariance (or lack thereof) in the one dune system. To achieve this, a constant $\tau_1$ value was chosen ($\tau_1 = 100$), and the simulation was run for dunes of varied areas: 6000 $l_0^2$ and 10000$l_0^2$. Again, the locations of the maximum surface shear stress and saturated flux were plotted Figure 11.

The aspect ratio, defined as the ratio of dune height, $H$, to dune length, $L$, is a commonly used metric for characterizing geometry in dune studies, including Jarvis et al. (2023). Figure 11 compares two dunes with different initial cross sectional areas subjected to identical flow conditions. To assess whether these dunes exhibit geometric scaling invariance, their aspect ratios were calculated and compared.

The topography shown in Figure 11 is a spatial average over 50 simulations. This

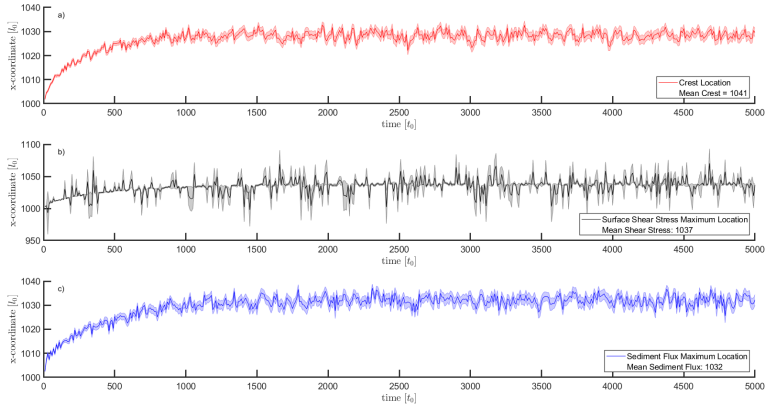



**Figure 12:** Maximum locations in x of the a) crest, b) shear stress maximum, and c) sediment flux maximum. Error bars are computed across 50 runs for a dune of area 6000 $l_0$ under flow conditions of $\tau_1 = 100$.

average was produced by recentering the dune position during the simulation on a timescale shorter than the data sampling interval. The method allowed for geometric features to be retained while smoothing the discrete structure of the model output. The resulting aspect ratios are:

- **Area 6000** $l_0$**:** 0.1954
- **Area 10000** $l_0$**:** 0.1939

The similarity of these variables indicate that the two dunes are approximately geometrically similar under the conditions considered. However, Kroy et al. (2002) proposed that small dunes do not obey strict scaling invariance. Their work evaluated dune sizes through the ratio $L/l_{sat}$, where $l_{sat}$ is the saturation length, and suggests that scale invariance is only exhibited in the limit as $L/l_{sat}$ approaches infinity. Under this assumption, neither dune shown in Figure 11 would be considered large enough to fully satisfy the assumptions scaling invariance.

**Error Analysis of Maximums:** Figure 12 shows the standard error across fifty simulations for each of the plotted maximums in Figure 8 and Figure 9. The oscillatory nature of each plot makes it difficult to evaluate the relative locations of each maximum at the timestep level. Figure 12 can also be used to validate the steady state claim. Each plot reaches a horizontal mean around when steady state is reached.



**Figure 13:** An example setup. A smaller upwind dune is chasing a larger downwind dune. Flow is flowing left to right, and the domain is periodic. The top frame displays the sediment profile, and the bottom shows the fluid behavior.

# 3 Two Dune System

#### 3.0.1 Dune Collisions

When individual dunes in a field migrate at different speeds, the dune pattern not only translates, but can also evolve temporally Bacik et al. (2020). In two dimensions, coalescence and ejection are the main mass-exchange collisions observed Diniega et al. (2010); Jarvis et al. (2023). Non-mass exchange interactions have been observed in nature Vermeesch (2011), as well as studied experimentally Bacik et al. (2020) . Bacik, among others, theorize that dune-dune repulsion occurs due to the separation bubble and subsequent wake shed from the lee slope of a dune. The increased turbulence experienced by a downwind dune of a similar size increases sediment flux and subsequently, the dune migration rate. Bacik et al. (2020).

# 4 Nonidentical Dunes

## 4.1 Effect of Flow speed on collision type

#### Dune fields are never stable

Over time, dune fields tend to evolve toward larger and more widely spaced dunes, making dune spacing a useful indicator for dune field maturity (Day and Kocurek, 2018). Additionally, as dune fields coarsen, the randomness and imperfections of nature create some variability as well. However, Day and Kocurek (2018) demonstrated that despite this growth, the non-dimensional spatial distribution of interactions remained constant, suggesting a fundamental link between spatial and temporal scales. This result further implies that in-

teractions persist even as dune fields mature. Earlier work had proposed the idea of a "perfect" dune field, in which equally sized and evenly spaced dunes would not interact. Field observations, however, have challenged this idea. For instance, Werner and Kocurek (1999) examined 16 dune fields– including the Rub 'al Khali Desert in the Arabian Peninsula–and found that natural dune fields seldom exhibit such a "perfect" arrangement.

A dune field responds to key morphological controls, primarily wind speed and sediment availability. The inherently dynamic nature of these controls help explain why a "perfect" dune field pattern is unlikely to occur in natural systems. From a modeling viewpoint, these drivers can be treated as boundary conditions, making it possible to investigate how the system responds to external forcing. Marvin et al. (2023) applied the dimensional interaction density described by Day and Kocurek (2018) to infer the timing of past shifts in morphological drivers. In their study, an interaction was identified when two dune crests occurred within a prescribed distance threshold. Satellite images yield discrete "snapshots" of an environment. As a result, distinguishing between interaction types becomes a challenge across a large number of dune fields, due to the lack of dynamic information.

Distinct interaction types may have different implications for the stability of a dune field, at least locally. Pattern coarsening–which forms a field of larger and more spaced out dunes–is considered stabilizing Jarvis et al. (2022). Coarsening is achieved through constructive interactions, such as coalescence. Ejections, (and splitting in 3 dimensions) may be regarded as non-stabilizing: the ejected bedform migrates downwind to initiate future interactions Gao et al. (2015b). Previous work Jarvis et al. (2023) found ejection and coalescence to occur with even frequency. However, if one collision type consistently dominates, it may influence the long-term stability of a dune-field.

Accurate modeling of long term dune field behavior requires an understanding of dune-dune interactions. This work focuses specifically on interactions between barchan dunes. Dune size is inversely related migration velocity Bagnold (1941), such that smaller dunes generally migrate faster than larger dunes, as demonstrated in Section 2. These differences in migration rate will inevitably lead to interactions over time. Jarvis et al. (2023) investigated the probabilities of coalescence and ejection as a function of dune sizes. In a $2D$ domain, these interactions were analyzed using the cross sectional areas of upwind and downwind dunes. Jarvis et al. (2023) explored how the cross-sectional area ratio, $r = \frac{A_u}{A_d}$, influences the eventual collision outcome, namely, coalescence or ejection. In all cases, the upstream dune area, $A_u$, is smaller than the downstream dune area, $A_d$, implying that $r < 1$. The three elementary types of dune interactions in $2D$ are described below.

**Two different types of coalescence** In downstream-dominant coalescence, the upstream dune climbs the stoss side of the downstream dune, and some of its sediment is deposited on the downwind lee side. The trough height between the two dunes increases, and eventually surpasses the height of the upstream peak. and some of its sediment is deposited on the lee side. Eventually, the upstream peak vanishes Gao et al. (2015a); Jarvis et al. (2023). Jarvis et al. (2023) observed this type of coalescence when the cross-sectional area ratio, $r$, was the smallest.

Upstream-dominant coalescence prevails when $r$ is increased compared to downstream-dominant events. The upstream dune once again climbs the stoss face of the downstream dune, but preserves its avalanching side and recirculation zone. The downstream dune is now in the recirculation zone of the dominant upstream heap. Avalanching sediment over the upstream lee side is trapped in the trough, where the shear stress is lowered. A flat plateau forms between the two dunes, and the now small downstream mass begins accelerating away, but is eventually overtaken by the upstream dune Gao et al. (2015a); Jarvis et al. (2023).

**Ejection** Ejection is observed when r is increased from its upstream-dominant coalescence value, while still remaining below 1 Jarvis et al. (2023). As the upstream dune begins climbing up the stoss face of the downstream heap, the bed shear stress on the stoss slope and in the trough remain sufficiently high to continue eroding Gao et al. (2015a). The downstream dune manages to escape the recirculation zone of the upstream dune.

A key result from the group's work is shown in Figure 16. Jarvis et al. (2023) used ReSCAL, in which flow strength is controlled through a prescribed critical shear stress threshold, $\tau_1$. Equation (4) describes the relationship between $\tau_1$ and the resulting velocity. In their study, $\tau_1$ was set to zero, meaning any wind would generate sand transport. As $\tau_1$ increases, the threshold for grain motion rises, as illustrated in Figure 15. While flow speed is known to be a dominant control on large-scale dune field evolution, its influence on individual dune interactions remains ambiguous. Near the transport threshold, sediment transport processes can vary subtly, and dune fields worldwide experience diverse wind regimes that warrant targeted investigation. The following section explores the effects of increasing $\tau_1$, which reduces flow velocity and overall sediment transport.

**In depth examination of past study** The cellular automaton model, ReSCAL (Section 1.2) was run over a periodic domain, in which the smaller upwind dune progressively caught up to and eventually collided with the downwind dune Figure 13.

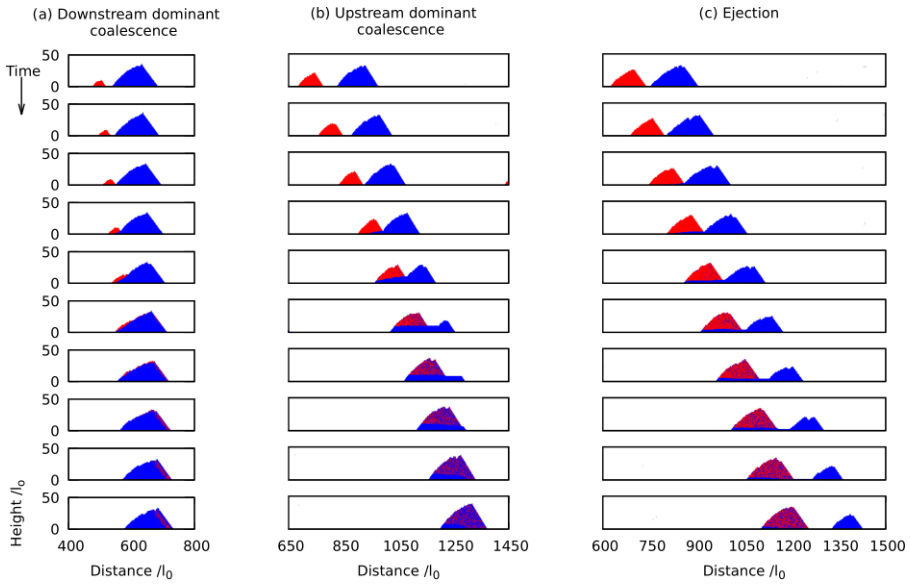



**Figure 14:** Figure adapted from Jarvis et al. (2023), showing the evolution of three principle $2D$ dune collisions.

Figure 16 summarizes the group's findings. When the two dunes are closer in size, more ejection cases occur. As $r$ decreases, dunes begin to coalesce, until all interactions are constructive in nature.

**Methods used in my experiment** ReSCAL was run with parameter inputs identical to Jarvis et al. (2023). The critical shear stress, $\tau_1$ was the only variable changed. Table 5 lists the parameters used in my simulations.

Each point in Figure 17 represents 150 - 200 simulations, spanning four different absolute dune areas. For each simulation, the collision outcome–either ejection or coalescence–was recorded, and the percent ejection was plotted.

$$f = 0.5 \cdot \tanh\left(\mathbf{b}x + \mathbf{c}\right) \tag{16}$$

### 4.1.1 Results

The plots in Figure 17 indicate that flow velocity exerts a measurable influence on collision outcome. Relative to the $\tau_1 = 0$ case, the tanh fit shifts to the right, suggesting that the transition zone from coalescence to ejection occurs at a larger area ratio. The fitted slope remains nearly unchanged between the

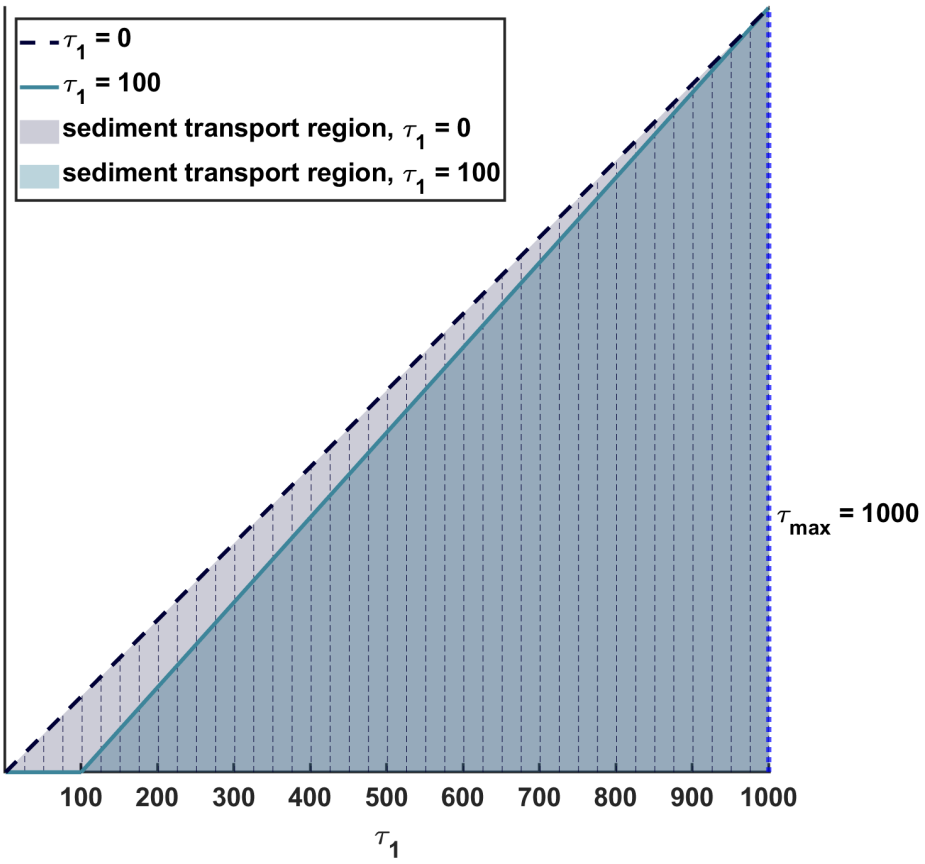



**Figure 15:** The effect of varying the critical shear stress ($\tau_{min}$ or $\tau_1$. Any shear stress value above the line will not yield sediment transport, while it will above the line. Therefore, increasing $\tau_1$ corresponds to decreased sediment transport, which mimics a slower flow velocity.

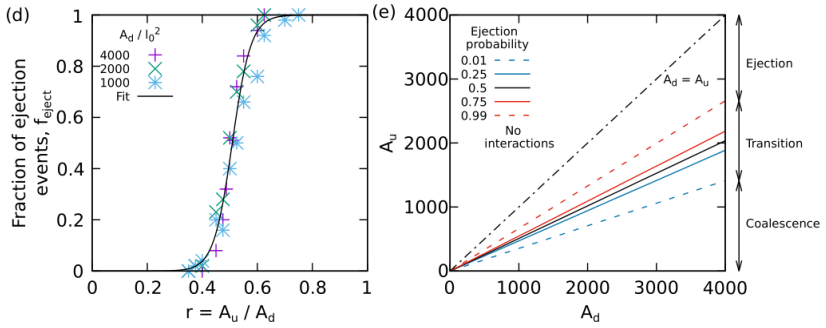



**Figure 16:** (d) Fraction of ejection events $f_{eject}$ as a function of area ratio $r = \frac{A_u}{A_d}$ and (e) Contours of probability of ejection as a function of $r$. Figure adapted from Jarvis et al. (2023). Each point represents a result from 50 simulations.

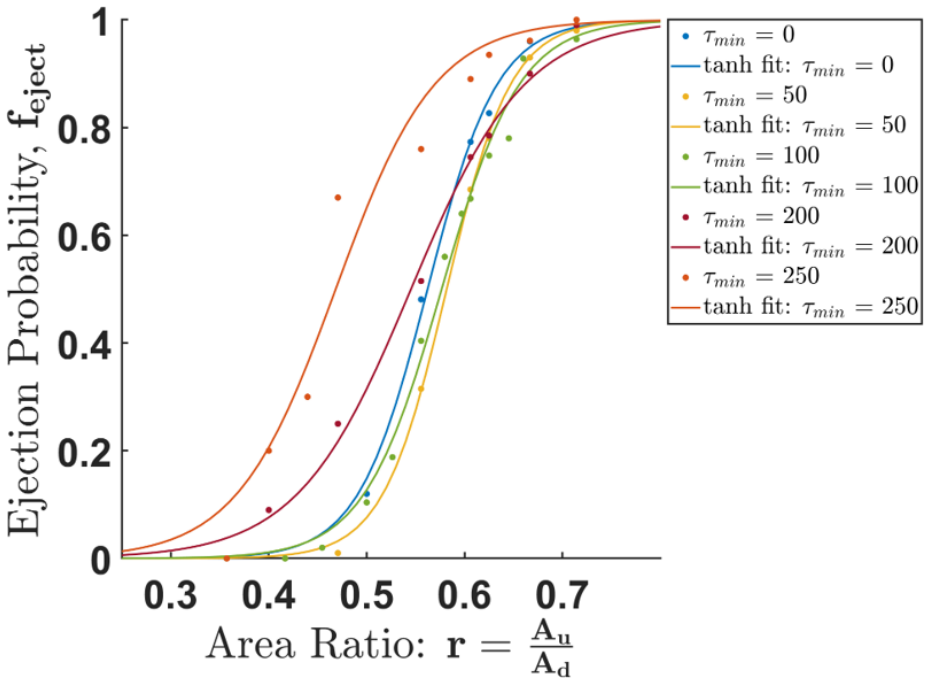



**Figure 17:** Ejection probabilities of dunes as a function of area ratio, $r$, under different flow conditions. The tanh fits are set by (16). All results are from my own data.

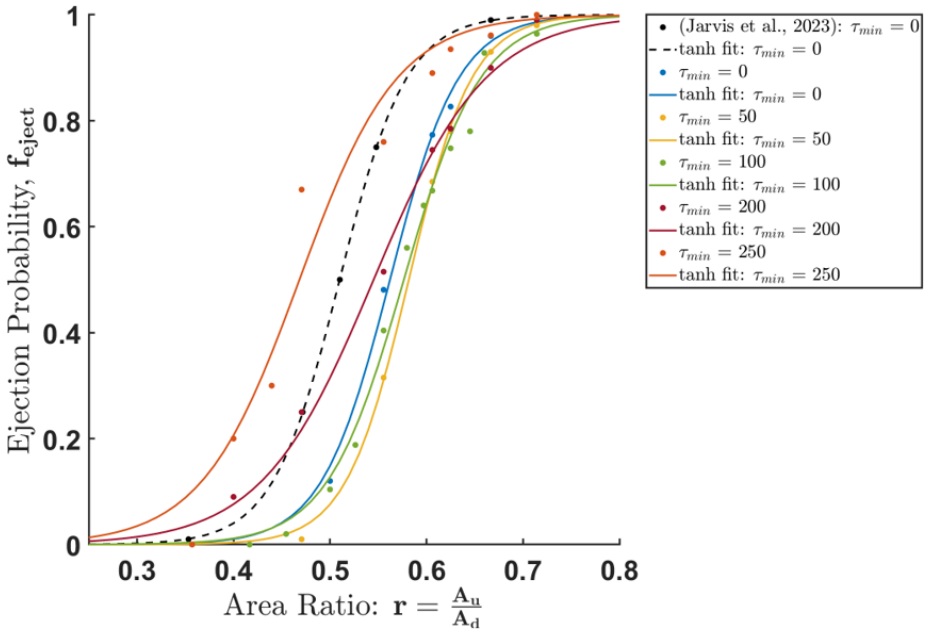



**Figure 18:** Ejection probabilities of dunes as a function of area ratio, $r$, under different flow conditions. Data from Jarvis et al. (2023) is plotted in black. The flow conditions of the yellow line are set by $\tau_1 = 0$, the same value used in Jarvis et al. (2023).

**Table 5:** Parameter values used in the model. Values adapted from Jarvis et al. (2023), with only $\tau_1$ changed.

| Symbol | Description | Model value |
|---|---|---|
| $l_0$ | length scale (i.e. cell size) | – |
| $t_0$ | timescale | – |
| $H$ | domain height | $300l_0$ |
| $L$ | domain length | $2000l_0$ |
| $\Lambda_0$ | transition rate for erosion | $4t_0^{-1}$ |
| $\Lambda_c$ | transition rate for deposition | $2t_0^{-1}$ |
| $\Lambda_t$ | transition rate for transport | $6t_0^{-1}$ |
| $\Lambda_g$ | transition rate for gravitational settling | $1000t_0^{-1}$ |
| $\Lambda_d$ | transition rate for diffusion | $0.02t_0^{-1}$ |
| $\Lambda_a$ | transition rate for avalanching | $10t_0^{-1}$ |
| $\theta$ | angle of repose | $35°$ |
| $\tau_1$ | critical shear stress for motion | 100 |
| $\tau_2$ | adjustable parameter | $1000\tau_0$ |

$\tau_1 = 0$ and $\tau_1 = 50$ cases. For $\tau_1 = 100$, however, the transition becomes slightly more gradual, even through the underlying data points do not differ substantially. Although some visual differences are apparent within the faster-flow regime, a statistical analysis similar to the one presented in Section 4.1.2 indicates no significant differences among the $\tau_1$ values associated with faster flows.

**A longer transition zone for slower flow regimes:** The $\tau_1 = 200$ and $\tau_1 = 250$ cases exhibit a clear divergence from the other conditions. A leftward shift is seen in the $\tau_1 = 200$ data, indicating an overall higher percentage of ejection outcomes, coupled with an earlier transition into the ejection regime. The trend continues even more severely for $\tau_1 = 250$. The shallower slope of the tanh fits in these two cases indicates a longer period of transition between coalescence and ejection. In other words, the relative range of area ratios where either collision outcome is likely to occur, is wider.

**Insight into the transition regime** Within the transition period from ejection to coalescence, ejected masses decrease in average size. Diniega et al. (2010) described the *extreme ejection* case, where a small heap manages to escape the downwind dune, but is not large enough to remain stable. The extended transition period in the slow flow region suggests that small ejections are more likely to be observed in areas with lower wind speeds.
In aeolian systems, small grains are the first to move, saltating close to the surface Dominguez et al. (2020). As the particles fall back to the bed, their impact mobilizes other particles. Near the shear stress threshold, impact driven

entrainment plays a larger role in particle motion; in contrast, at higher flows, the bulk flow becomes the dominant mechanism.
ReSCAL, as a cellular automaton model, does not directly incorporate the sporadic nature of near threshold particle motion. However, as $\tau_1$ increases (flow velocity decreases), the sediment flux, $Q$, declines significantly Figure 5. In the simulations, this behavior translates to a smaller percentage of cells being mobilized. Through transitions, a doublet will still affect a neighboring cells, but the propagation of that change in model state will spread more slowly throughout the cellular space. This results in the longer transition period, but does not necessarily reflect the physics of slow flow.

### 4.1.2 Examining the Statistical Significance of Slow and Fast Flow Regimes

Visual inspection of Figure 18 suggests that the $\tau_1 = 200$ and $\tau_1 = 250$ cases exhibit distinct behavior from the of $\tau_1 = 0$, $\tau_1 = 50$, $\tau_1 = 100$ cases and the Jarvis et al. (2023) data. This distinction is physically relevant because wind regimes in natural dune fields varying both strength and persistence, which in turn influences sediment transport and dune migration rates. As a result, identifying whether collision outcomes organize into distinct flow regimes is important for our understanding of how dune interactions may differ across natural environments.

To move beyond qualitative comparison, I assessed whether the differences between tanh fits were statistically meaningful. Two apparent flow regimes emerge in Figure 18: a slower regime and a faster regime, within which the fitted curves behave similarly. I therefore selected the tanh fits in (16) for $\tau_1 = 250$ and $\tau_1 = 50$ as representative of the slower and faster regimes. Because each fit is based on a relatively small number of data points, conventional statistical approaches such as hypothesis testing or bootstrapping may yield unreliable results.

Table 6 therefore summarizes the fitted coefficients for each selected case, with the following interpretations:

- The parameter $b$ describes the steepness of the curve, or the range of area ratios in the transition zone. The value of $b$ in the $\tau_1 = 50$ instance is greater (15.306) compared to (9.863) of the $\tau_1 = 250$ instance. Their respective 95% confidence intervals overlap slightly, yet one could choose to note that their 94% confidence intervals do intersect. A reasonable conclusion can be drawn: under faster flow conditions, a smaller range of area ratios, $r$, fall into the zone where both ejection and coalescence outcomes are likely.
- The observable rightward shift of the faster flow cases is reflected in $c$,

| | Steepness $b$ | 95% CI | Shift $c$ | 95% CI | Midpoint $x_0$ | Approx. Range |
|---|---|---|---|---|---|---|
| $\tau_1 = 50$ | 15.306 | (14.047, 16.564) | -8.903 | (-9.651, -8.155) | 0.582 | (0.581, 0.581) |
| $\tau_1 = 250$ | 9.863 | (5.393, 14.332) | -4.619 | (-6.678, -2.560) | 0.468 | (0.466, 0.475) |
| Difference | **5.443** | | **-4.284** | | **0.113** | |

**Table 6:** Comparison of tanh fit parameters for different $\tau_1$ values. Midpoint $x_0 = -c/b$. Approximate ranges are derived from 95% confidence intervals.

which equals (-8.903) and (-4.619) for $\tau_1 = 50$ and $\tau_1 = 250$ respectively. The 95% confidence intervals for each value do not overlap. These results indicate that ejection is more prevalent in the slow flow regime.

- The midpoint, $x_0$, is approximated as $x_0 = -c/b$ and reflects the inflection point of each function. When $x < x_0$ the probability of coalescence is greater, at $x = x_0$ coalescence and ejection outcomes are equally likely, and if $x > x_0$ ejections occur more frequently. The midpoint of $\tau_1 = 50$ ($x_{0,50} = 0.582$) is greater than ($x_{0,250} = 0.468$). The approximate ranges, calculated using the 95% intervals, do not overlap. Fast flow conditions shift the transition into the ejection zone toward larger area ratios.

#### 4.1.3 Discussion

The above plot reveals several important trends. Under slower flow regimes, the transition from coalescence to ejection occurs at smaller area ratios. In addition, the transition zone–defined as the range of area ratios that produce both coalescence and ejection outcomes–is broader for the slower flow cases of $\tau_1 = 200$ and $\tau_1 = 250$. By contrast, the faster flow cases ( $\tau_1 = 0$, $\tau_1 = 50$, and $\tau_1 = 100$), do not differ substantially from one another in either slope or inflection point. These results suggest that dune interactions exhibit two distinct behavioral regimes associated with flow strength. Under lower-strength flow, dunes transition from coalescence to ejection at smaller cross sectional area ratios (or relative masses). Under faster flow conditions, coalescence persists to larger area ratios, after which the transition to ejection occurs more rapidly.

It is important to recall that coalescence can occur in two forms: downwind-dominant coalescence and upstream-dominant coalescence. Of these, upstream-dominant coalescence is more dynamically similar to ejection, as both outcomes involve growth of the upstream dune at the expense of the downstream dune. The wider transition zone observed under slower flow conditions raises an important question: does weaker flow increase the likelihood of upstream-dominant

coalescence, thereby broadening the range of conditions over which the system exhibits mixed behavior?

At the beginning of this section, the ongoing debate regarding dune-field stability was introduced. Although dune fields are continually evolving, certain processes have a stabilizing effect, one of the most significant being pattern coarsening. Constructive interactions, such as coalescence, contribute positively to this process. As shown in Figure 17, the faster flow cases transition to ejection at larger area ratios and therefore experience a broader range of coalescence outcomes. This may suggest that dune fields under faster flow conditions are more stable. Simultaneously, it is important to consider the opposite argument, that slower flow cases enhance dune-field stability. Individual interaction timescales are longer under slower flows, but that alone does not necessarily imply greater dune-field stability. The internal structure of these interactions–specifically whether they proceed through down-wind dominant or upwind-dominant coalescence–may be equally important.

It should be noted that the experiments described above were conducted in a two-dimensional domain, whereas turbulence and flow behavior are inherently three-dimensional. For example, in barchan dunes, much of the sediment is lost from the horns, and a significant portion of the flow capable of transporting grains is directed outward toward them. The two-dimensional case considered here may therefore provide insight into sediment and flow interaction over the crest, but it cannot capture the full dynamics of the natural system. Nevertheless, within this simplified framework, the distinction between modes of coalescence provides a way to interpret why the transition from coalescence to ejection depends so strongly on flow strength. The following section therefore focuses on upstream-dominant coalescence as a potential mechanistic bridge between two diverse collision outcomes.

# 5 Mid-Interaction dynamics - will the dune eject?

### 5.0.1 An overview

If sufficiently large, barchan dunes will affect the fluid moving above them. As water in subaqueous settings, or air in aeolian settings, flows over a barchan barchan, it flow separates at the brink point, producing a recirculation bubble on the lee side (Bagnold, 1941). A turbulent wake shed by the upstream dune can increase turbulence over its downstream neighbor (Bacik et al., 2020; Bristow et al., 2018, 2020, 2019). Flow separation occurs on lee slopes with angle greater than 10° (Lefebvre and Cisneros, 2023); the dunes examined in this work have an angle of repose 35°. A reasonable claim can be made that a recirculation bubble will always form on the lee side of the upstream dune, and play a role

in any type of interaction if the dunes are sufficiently close.

**Non-mass exchange interactions** Non-mass exchange interactions have been observed in nature (Vermeesch, 2011), as well as studied experimentally (Bacik et al., 2020). Bacik, among others, theorized that dune-dune repulsion results from the wake shed by the upstream dune. Increased turbulence over a similarly sized downstream dune enhances sediment flux, and in turn, dune migration (Bacik et al., 2020).

Our group has attempted to model these dune "repulsions" in the past: Karol Bacik conducted subaqueous experiments in the quasi-2D inifite domain of an annular flume. Two piles of glass beads of identical mass were initially separated by $\pi/4$ in the azimuthal direction. Upon the initiation of motion, the heaps quickly formed into dune shapes, and began migrating. Although their masses were equal, their observed migration velocities differed. The upstream dune maintained a close to constant speed while the downstream dune sped away until the dunes converged to an antipodal configuration and held the same constant migration rate. As flow travels over the crest of the upwind dune, it separates, eventually reconnecting on the lee side. The dune-dune feedback is closely related to the turbulent structures shed in the wake of the upwind dune. Bacik et al. (2020) observes sudden bursts of sediment flux in the downstream dune which increase the migration velocity of the downstream dune. The experiment of Bacik et al. (2020) motivated my attempt to recreate dune-dune repulsion in ReSCAL.

**An unsuccessful attempt at reproducing dune-dune repulsions** In two-dimensions, dunes did not exhibit repulsion, likely due to the three-dimensional nature of turbulent flow. While I was unable to reproduce the dune-dune repulsion reported by Bacik et al. (2020), a model test was conducted to verify that the two-dune configuration behaved physically. Dunes of identical size and shape were positioned on opposing sides of a period domain under the same flow conditions.

Dune migration is primarily controlled by two factors: flow speed and dune size. Consequently, dunes of equal size are expected to migrate at the same rate. Figure 19 illustrates the initial domain setup. Over the course of the simulation, the dunes migrated slightly closer together, which can be attributed to simulation noise and random loss of sediment. The observed shift was minor–approximately 30 $l_0$ within a 2000 $l_0$ domain–and occurred over one million timesteps ($1,000,000\, t_0$).

Having confirmed that the basic system behaved as expected, I proceeded to investigate non-identical dunes.



**Figure 19:** Two identical dunes of area $2000l_0^2$, initiated at a distance of $1000l_0$, in a domain of length $2000\, l_0$. The top portion displays the sediment profile, and the bottom half shows the fluid behavior.

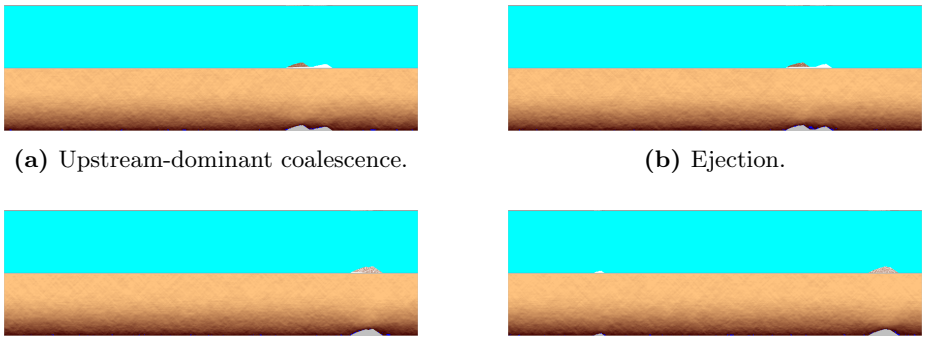
### 5.0.2 Exploring the effect of the recirculation bubble on mass-exchange interactions

For two identical dunes, the migration rates predicted from mass alone are theoretically the same, isolating the influence of the recirculation bubble. For dunes of unequal size, however, the effect of mass–represented in this study by the cross sectional area– must be considered in addition to the recirculation bubble. In all mass-exchange collisions, the upstream dune alters the incoming flow and thus affects the downstream dune throughout the interaction. The magnitude of this contribution to the downstream dune's velocity will vary.

In $2D$ ejection collisions, a smaller dune manages to break off from the larger downwind dune and accelerate away (Jarvis et al., 2022; Diniega et al., 2010). Jarvis et al. (2023) showed that the area ratio between the upwind and downwind dunes is a probabilistic variable in the collision outcome. The question of whether a dune will successfully eject - under a set of known conditions- remains open ended.

The transition region between coalescence and ejection may provide some insight into the mechanics of ejection. Because the model is probabilistic, there exists a range of area ratios for which both coalescence and ejection are observed. Within this range, the two outcomes are effectively in competition, suggesting that small differences in the interaction dynamics can determine the final result. A smaller upstream dune will inevitably migrate into a larger dune, as shown by Figure 14, and the early stages of upstream-dominant coalescence and ejection can appear quite similar Figure 20. In both cases, the upstream dune approaches and begins to interact with the downstream dune while maintaining its crest and lee-side recirculation zone.

The key difference lies in the evolution of the inter-dune region. In upstream-dominant coalescence, this region develops into a broad plateau that persists for



**(a)** Upstream-dominant coalescence.

**(b)** Ejection.

**Figure 20:** Upstream-dominant coalescence is displayed on the left, and ejection is shown on the right. A mid-interaction point is shown in the top panels, and outcomes are displayed in the bottom.

some time, indicating that sediment deposition into the trough outpaces erosion out. By contrast, in classical ejection–when $r$ approaches 1–the inter-dune region is eroded more effectively, allowing the downstream feature to detach and persist as an independent dune. Further into the transition region, however, ejection may occur when when a flat plateau exists for the majority of the interaction.

One might expect the larger upstream dune in ejection cases to produce a stronger or more extensive flow separation bubble, thereby further suppressing sediment transport. While this may be true, the results suggest that the size of the recirculation zone alone does not determine the outcome. In upstream-dominant coalescence, the upstream dune appears to erode downstream stoss side more rapidly–and therefore more rapidly supply sediment to the inter-dune region–than in ejection cases. As a result, a plateau forms and is maintained because deposition into the trough exceeds the rate at which the region is eroded away. This highlights the importance of considering not only the flow separation zone, but also the relative migration rates of the upstream and downstream dunes.

In this section, ReSCAL is used to further investigate mid-interaction mechanics to better understand why ejection occurs. Diniega et al. (2010) describes an extreme case of ejection, where a downwind hump manages to separate but is too small to remain stable and eventually disappears. Smaller ejection cases may therefore represent an intermediate state between extreme ejection events, which occur near an area ratio of $r = 1$, and complete coalescence.

### 5.0.3 First Attempt

**Methods** Two $2D$ dunes were initially spaced at a distance $\lambda$ apart, with the larger dune located upstream. At first glance, this configuration may appear to contradict the commonly accepted ideas that smaller dunes migrate faster than larger dunes Bagnold (1941). However, the purpose of this was not to reproduce a naturally evolving dune pair, but rather to isolate and mimic key stages of the ejection process. Another physical way for such configuration to occur is as a transient product of changing flow conditions, which may temporarily alter migration rates, sediment supply, or dune geometry in a way that makes these interactions possible.

Because sediment flux is directly related to the local flow strength, I tracked the location and magnitude of the maximum sediment flux on both the upwind and downwind dunes. I expected that, as the upstream dune began to influence the downstream dune, the sediment-flux signal on the downstream dune would reflect the onset and evolution of that interaction.

By varying the initial separation distance between the two dunes, Figure 22 was produced. The ratios of $Q_{max}$ values are plotted, with each line representing an average of 10 simulation runs. The results show two distinct behaviors: some curves decay towards zero (or a very small value), while others approach a non-zero plateau. Cases in which the ratio tends toward zero correspond to coalescence, indicating that the downstream dune is eventually absorbed. In contrast, cases that level off at a finite value correspond to ejection, indicating that the downstream dune persists as a distinct feature.

A transition between two outcomes appears to occur between $\lambda = 230\, l_0$ and $\lambda = 240\, l_0$. The simulations shown in this figure were based on the initial configuration displayed in Figure 21, in which the cross-sectional area of the downstream mass–intended to represent the eventual ejected dune–was set to 0.2 times that of the upstream dune. Relative to many ejection events observed in the study, this downstream dune is comparatively large. As a result, this experiment primarily represents cases with an area ratio closer to $r = 1$, rather than the broader transition regime of interest.

Although this initial configuration was ultimately not well suited for studying the full transition regime, it was still useful for identifying a critical separation distance, $\lambda \approx 235\, l_0$, at which interaction behavior changes. More importantly, this value marks the point beyond which the model begins to lose physical relevance for ejection events.

In future analysis, sediment flux was retained as a diagnostic for quantifying how the upstream dune influences the downstream dune, but later analyses were designed around a more representative (i.e. smaller) "final form" or the ejected mass–one that better reflects the types of interactions expected within



**Figure 21:** Upstream dune of area $A_u = 10,000\, l_0^2$, and downstream dune of area $A_d = 2000\, l_0^2$. The initial separation, $\lambda = 230\, l_0$.

the transition regime.

**The transition zone** I selected the base case of $\tau_1 = 0$ to investigate the whether two-dune interactions result in ejection or coalescence. During ejection, both the upwind and downwind dunes retain distinct crests, even as their relative sizes reverse. These crest positions can therefore be tracked through time, allowing identification of the minimum separation distance. The minimum value was chosen at a stage when the downwind dune had reached a stable shape; some interaction stages produce a plateau with multiple potential crest locations Figure 23a.

Motivated by the results of Figure 23, I used the area of the ejected heap, $247\, l_0^2$, together with the area of the larger dune, $14596\, l_0^2$, to investigate the flipped configuration. In this case, the upwind dune was initiated with an area of $A_u = 14596\, l_0^2$ and the downwind dune with an area of $A_d = 237\, l_0^2$, separated by a distance of $165\, l_0$. Across 10 simulations, the downstream dune escaped in 100% of the cases.

I then increased the size ratio by a factor of two to test whether the interaction outcome could be shifted towards merging. In this case, the downwind dune did not escape; instead, the dunes coalesced into a single dune Figure 25.

Since the sediment flux is driven by flow strength, observing the sediment flux gives insight into the flow behavior. The sediment flux ratios can show the effect an upwind dune has on its downstream neighbor, as displayed in Figure 26. When $A_d = 247\, l_0^2$–which results in a successful ejection– the maximum sediment flux ratio, $Q_{max,u}/Q_{max,d}$ plateaus. However, when $A_d = 500\, l_0^2$, the sediment flux ratio decreases significantly and approaches a value very close to zero. At the point where the lines cross, merging is in process in the $A_d = 500 l_0^2$ case, about 50% of the original downwind dune area has coalesced.

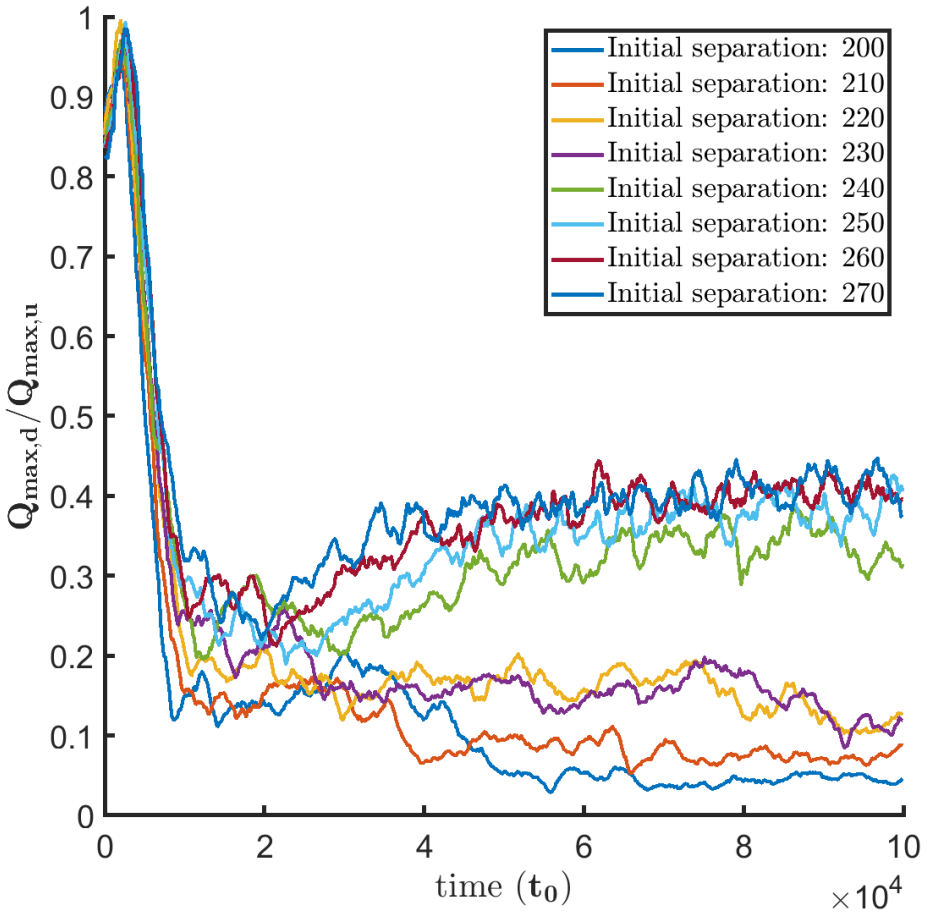



**Figure 22:** Ratios of the maximum sediment flux of the downwind dune over the upwind dune.



**(a)** A picture mid ejection, where multiple crests can be seen on the downwind dune.

**(b)** The downwind dune formed into a more stable shape, 40,000 $t_0$ after the previous snapshot.

**(c)** The ejected dune appears on the left size of the periodic domain at $t = 300,000\, t_0$.

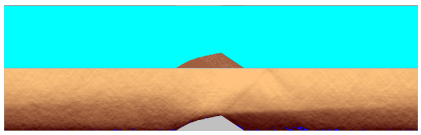
**Figure 23:** The dunes were initiated with $A_u = 5,5555\, l_0^2$ and $A_d = 10,000\, l_0^2$. The ejected dune in panel c) has an area of $247\, l_0^2$, and the larger dune grows to $14596\, l_0^2$.



**Figure 24:** Initial dune setup: $A_u = 14596\, l_0^2$, and $A_d = 237\, l_0^2$.



**(a)** Initial dune setup: $A_u = 14596\, l_0^2$, and $A_d = 500\, l_0^2$.



**(b)** Initial dune setup: $A_u = 14596\, l_0^2$, and $A_d = 500\, l_0^2$.

**Figure 25:** When the downstream dune area was increased by $\tilde{1}00\%$ from Figure 24, ejection did not successfully occur.

Figure 27 was produced using the same initial configuration described in Figure 23. In this setup, a smaller upstream dune with cross-sectional area $A_u = 5,555\, l_0^2$ interacts with a larger dune with area $A_d = 10,000\, l_0^2$, corresponding to an area ratio of $r \approx 0.55$. Ten simulations were performed for this configuration, two of which are shown in Figure 27. Under flow conditions of $\tau_1 = 0$, the ejection probability for the specific area ratio is approximately 50%, consistent with the outcomes observed in these simulations. In cases where ejection does not occur, the interaction instead proceeds through upstream-dominant coalescence.

An important distinction between the two outcomes is their characteristic times. Across all ten simulations, the upstream-dominant coalescence cases resolve more quickly than the ejection cases. The result is somewhat counterintuitive, since in ejection the upstream dune does not fully absorb the downstream mass. However, the difference appears to arise from the mechanics of sediment exchange in the inter-dune region. In upstream-dominant coalescence, sediment in the trough is not eroded away efficiently, and the upstream dune primarily migrates over and incorporates the material. In ejection, by contract, sediment continues to be eroded from the trough, prolonging the interaction and allowing the downstream mass to detach and reorganize into a separate dune.

A useful continuation of this analysis would be to examine the timing and magnitude of the sediment flux drop, as well as the length of the plateau in the two

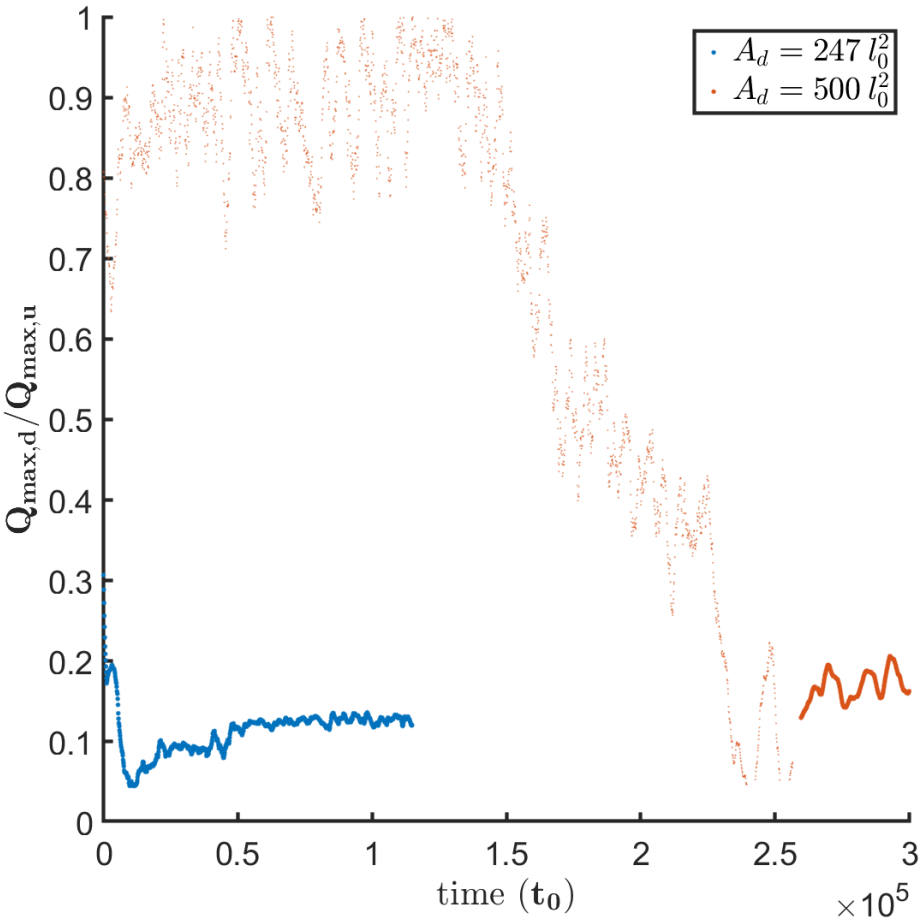



**Figure 26:** Ratio of maximum sediment fluxes for initial setup shown in Figure 24 and for the real ejection outcome shown in Figure 23 (orange). The darker orange line represents the point in which the ejected dune separates from the upwind, larger dune.

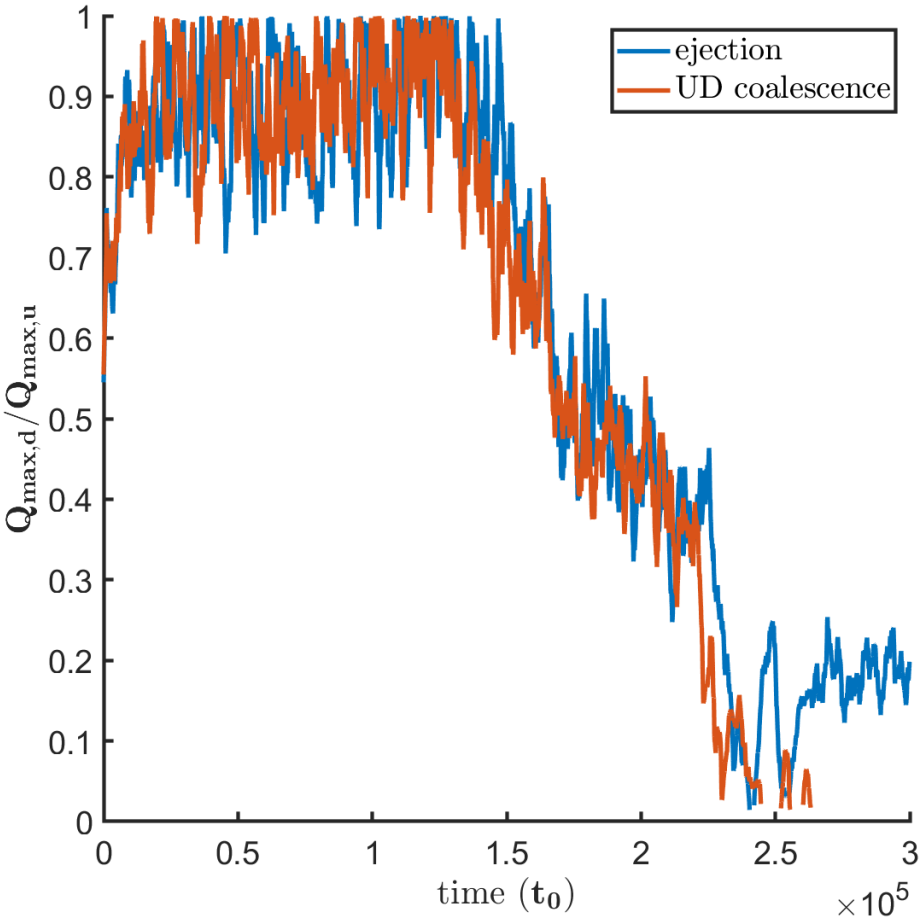



**Figure 27:** An upwind dune of area $A_u = 5555\, l_0^2$ collides with a downwind dune of area $A_d = 10000\, l_0^2$ in conditions of $\tau_1 = 0$, resulting in Ejection (blue) and Upstream-Dominant Coalescence (orange). Missing data points correspond to a plateau, where no clear sediment flux maximum can be found for the downstream mass.



**(a)** A plateau exists between the larger upwind bedform and much smaller downwind heap.



**(b)** A small dune still manages to eject at the end of the run.

**Figure 28**

cases. In the successful ejection case, a plateau is still present and is reflected in the blue curve as the sharp drop in the $Q_{max}$ ratio near $250,000\,t_0$.However, unlike in upstream-dominant coalescence, this plateau does not persist indefinitely, and the downstream mass ultimately reorganizes into an independent dune. This suggests that the presence of a plateau alone is not sufficient to determine the outcome;rather, its persistence and erosion timescale may be the more important controls.

**Limitations of ReSCAL** In ReSCAL, dunes are initialized as triangular piles and must evolve before reaching steady-state morphology. As a result, the downwind bedform experiences the wake shed by the upwind dune immediately, even though the upwind heap's initial shape is not yet representative of a natural dune. Consequently, this experiment is intended to provide mechanistic insight into the ejection process rather than to determine the precise separation distances or ejected dune area that mark the line between upsteam-dominant coalescence and ejection.

### 5.0.4 Future Work

A natural extension of this analysis is to non-dimensionalize the separation lengths associated with the interactions. The recirculation zone plays an important role in determining whether the downstream mass is preserved or absorbed. Thus, relating the separation length to dune geometry may provide a more systematic way to compare outcomes across varying flow conditions, and investigate scaling invariance in the context of collision type (in the transition zone).

As shown in Figure 17, slower flow regimes display a broader transition region between coalescence and ejection. Because upstream-dominant coalescence and ejection share similar dynamics during the middle stages of interaction, it would be informative to assess whether downstream-dominant coalescence can also occur for area ratios within this transition zone. Doing so could help determine whether the transition regime is controlled primarily by inter-dune trough evolution–if all coalescence cases are upstream-dominant–or instead by more general differences in flow structure.

### 5.0.5 Connection to results of Interaction Probabilities Experiment

A broad motivation for this thesis is to understand which types of dune interactions are favored under different flow regimes. This question has implications for interpreting dune-field evolution in both laboratory and natural settings, since the relative frequency of coalescence and ejection may influence whether a dune field tends toward patttern coarsening or preserves a wider distribution

of dune sizes.

As shown in Figure 5, sediment fluxes decrease significantly with increasing $\tau_1$. If the sediment flux over the downstream dune is strongly influenced by the wake generated by the upstream dune, then these changes in transport may help explain the collision trends shown in Figure 17. In particular, the earlier transition to ejection observed in slower flow cases ($\tau_1 = 200$ and $\tau_1 = 250$) may be related to differences in lee-side flow structure and the resulting ability of the downstream dune to sustain sediment transport.

# 6 Section 5: Conclusions

Dune-fields are inherently dynamic systems, as their two main morphological controls–flow strength and sediment availibility– are transient in nature. Despite this transience, certain processes can have a stabilizing effect on field-scale organization. Constructive collisions, such as coalescence, increase dune spacing and contribute to pattern coarsening, an accepted stabilizing mechanism.

My work uses the open-source cellular automaton model ReSCAL to investigate one of these key morphological controls: flow strength. I begin with a single dune configuration in which a triangular sand heap evolves into a barchan dune shape, before turning to the more complex and compelling two-dune problem. Because dunes migrate at different rates, collisions are an inevitable component of dune field evolution. This work focuses on two principle types of two-dimensional dune interactions: coalescence and ejection. Coalescence is further divided into upstream-dominant and downstream-dominant cases.

Natural dune fields provide clear evidence that flow strength can vary widely across environments. Dunes in Rub al Khali on the Arabian Peninsula can migrate 20–30 meters per year under persistent strong winds. In contrast, dunes in the Taklamakan Desert in China, where strong wind events are less frequent, may migrate at annual rates of only about 0.5 meters. The Kalahari Dunes of southern Africa, which experience a more moderate wind climate, typically migrate at rates of approximately 2 meters per year. These examples illustrate how dunes around the world are subjected to different wind regimes and, consequently, exhibit distinct patterns of evolution. As a result, flow strength is an important variable to consider when characterizing overall field behavior and the individual dune collisions that are the focus of this thesis.

Motivated by the work of Jarvis et al. (2023), who investigates the probabilistic nature of different dune interactions as a function of area ratio, I vary the flow strength and identify two distinct regimes: a faster-flow regime that exhibits behavior very similar to that described in Jarvis et al. (2023), and a slower regime with notably different collision behavior. In the slower-flow regime, the

transition from coalescence to ejection occurs at smaller area ratios. In addition, the transition zone–defined as the range of area ratios that produce both coalescence and ejection outcomes–is broader for the slower flow cases of $\tau_1 = 200$ and $\tau_1 = 250$.

This raises an interesting question: if stronger flow conditions produce more cases of coalescence, could dune fields subjected to stronger winds be more morphologically stable?

Inspired by the contrasting widths of the transition zones, the final portion of this thesis begins to investigate the regime in which both ejection and coalescence are possible. To better understand the mechanisms that may influence final collision outcome, I examine the the potential role of the recirculation bubble through analysis of the maximum sediment fluxes associated with both the upstream and downstream bedforms. While area ratio remains a primary control, the results suggest that within the transition zone of slower flows, flow behavior itself may exert a comparatively stronger influence on the eventual outcome than it does under faster-flow conditions. This points to the possibility that since flow is often turbulent and chaotic in nature, outcomes more controlled by the flow behavior as opposed to the area ratio may have reduced stability.

Overall, this work demonstrates that flow strength is not merely a background environmental or simulation parameter, but a fundamental control on dune field collision dynamics. By showing that changes in flow strength can shift collision thresholds, broaden transitional behavior, and potentially alter field-scale stability, this thesis highlights the importance of considering flow regimes alongside morphology when interpreting and modeling dune-field evolution.

# References


Alvarez, C. A., Lapôtre, M. G. A., Swann, C., Ewing, R. C., et al. (2025). Ripples formed in low-pressure wind tunnels suggest mars's large windblown ripples are not impact ripples. *Nature Communications*, 16:2945.

Andreotti, B., Claudin, P., and Pouliquen, O. (2002). Selection of dune shapes and velocities. *European Physical Journal B*, 28:321–339.

Andreotti, B., Forterre, Y., and Pouliquen, O. (2013). *Granular Media: Between Fluid and Solid*. Cambridge University Press, Cambridge, UK.

Ashkenazy, Y., Yizhaq, H., and Tsoar, H. (2012). Sand dune mobility under climate change in the kalahari and australian deserts. *Climatic Change*, 112(3):901–923.

Baas, A. C. W. and Delobel, L. A. (2022). Desert dunes transformed by end-of-century changes in wind climate. *Nature Climate Change*, 12(11):999–1006.

Bacik, K., Lovett, S., Caulfield, C.-c., and Vriend, N. (2020). Wake induced long range repulsion of aqueous dunes. *Physical Review Letters*, 124:054501.

Bagnold, R. A. (1941). *The Physics of Blown Sand and Desert Dunes*. CRC Press, Boca Raton, FL.

Best, J. (2005). The fluid dynamics of river dunes: a review and some future research directions. *Journal of Geophysical Research: Earth Surface*, 110(F04S02).

Binnig, G., Gerber, C., Stoll, E., Albrecht, T. R., and Quate, C. F. (1986). Atomic resolution with atomic force microscope. *Europhysics Letters (EPL)*, 2(4):257–263.

Bishop, M. A. (2007). Point pattern analysis of north polar crescentic dunes, mars: A geography of dune self-organization. *Icarus*, 191(1):151–157.

Bristow, N. R., Blois, G., Best, J. L., and Christensen, K. T. (2018). Turbulent flow structure associated with collision between laterally offset, fixed-bed barchan dunes. *Journal of Geophysical Research: Earth Surface*, 123(9):2157–2188.

Bristow, N. R., Blois, G., Best, J. L., and Christensen, K. T. (2019). Spatial scales of turbulent flow structures associated with interacting barchan dunes. *Journal of Geophysical Research: Earth Surface*, 124(5):1175–1200.

Bristow, N. R., Blois, G., Best, J. L., and Christensen, K. T. (2020). Secondary flows and vortex structure associated with isolated and interacting barchan dunes. *Journal of Geophysical Research: Earth Surface*, 125(2):e2019JF005257. e2019JF005257 10.1029/2019JF005257.

Bruno, L., Horvat, M., and Raffaele, L. (2018). Windblown sand along railway infrastructures: A review of challenges and mitigation measures. *Journal of Wind Engineering and Industrial Aerodynamics*, 177:340–365.

Charru, F., Andreotti, B., and Claudin, P. (2013). Sand ripples and dunes. *Annual Review of Fluid Mechanics*, 45(1):469–493.

Chopard, B. and Droz, M. (1998). *Cellular Automata Modeling of Physical Systems*. Collection Aléa-Saclay: Monographs and Texts in Statistical Physics. Cambridge University Press, Cambridge, UK.

Courrech du Pont, S., Rubin, D. M., Narteau, C., Lapôtre, M. G., Day, M., Claudin, P., Livingstone, I., Telfer, M. W., Radebaugh, J., Gadal, C., Gunn, A., Hesp, P. A., Carpy, S., Bristow, C. S., Baas, A. C., Ewing, R. C., and Wiggs, G. F. (2024). Complementary classifications of aeolian dunes based on morphology, dynamics, and fluid mechanics. *Earth-Science Reviews*, 255:104772.

Day, M. and Kocurek, G. (2018). Pattern similarity across planetary dune fields. *Geology*, 46(11):999–1002.

Diniega, S., Glasner, K., and Byrne, S. (2010). Long-time evolution of models of aeolian sand dune fields: Influence of dune formation and collision. *Geomorphology*, 121(1–2):55–68.

Diniega, S., Kreslavsky, M. A., Radebaugh, J., Silvestro, S., Telfer, M., and Tirsch, D. (2017). Our evolving understanding of aeolian bedforms, based on observation of dunes on different worlds. *Aeolian Research*, 26:5–27.

Dominguez, L., Bonadonna, C., Forte, P., Jarvis, P. A., Cioni, R., Mingari, L., Bran, D., and Panebianco, J. E. (2020). Aeolian remobilisation of the 2011-cordón caulle tephra-fallout deposit: Example of an important process in the life cycle of volcanic ash. *Frontiers in Earth Science*, 7:343.

Durán, O., Schwämmle, V., Lind, P. G., and Herrmann, H. J. (2011). Size distribution and structure of barchan dune fields. *Nonlinear Processes in Geophysics*, 18(4):455–467.

Durán, O., Claudin, P., and Andreotti, B. (2014). Direct numerical simulations of aeolian sand ripples. *Proceedings of the National Academy of Sciences*, 111(44).

Elbelrhiti, H., Claudin, P., and Andreotti, B. (2005). Field evidence for surface-wave-induced instability of sand dunes. *Nature*, 437:720–723.

Endo, N., Taniguchi, K., and Katsuki, A. (2004). Observation of the whole process of interaction between barchans by flume experiments. *Geophysical Research Letters*, 31(12).

Filhol, S. and Sturm, M. (2015). Snow bedforms: a review, new data, and a formation model. *Journal of Geophysical Research: Earth Surface*, 120(9):1645–1669.

Gao, X., Narteau, C., and Rozier, O. (2015a). Development and steady states of transverse dunes: A numerical analysis of dune pattern coarsening and giant dunes. *Journal of Geophysical Research: Earth Surface*, 120(10):2200–2219.

Gao, X., Narteau, C., Rozier, O., Zhang, D., and Courrech du Pont, S. (2015b). Phase diagrams of dune shape and orientation depending on sand availability. *Scientific Reports*, 5:14677.

Hersen, P., Andersen, K. H., Elbelrhiti, H., Andreotti, B., Claudin, P., and Douady, S. (2004). Corridors of barchan dunes: Stability and size selection. *Phys. Rev. E*, 69:011304.

Hersen, P. and Douady, S. (2005). Collision of barchan dunes as a mechanism of size regulation. *Geophysical Research Letters*, 32(21):L21403.

Hersen, P., Douady, S., and Andreotti, B. (2002). Relevant length scale of barchan dunes. *Phys. Rev. Lett.*, 89:264301.

Jarvis, P. A., Bacik, K. A., Narteau, C., and Vriend, N. M. (2022). Coarsening dynamics of 2d subaqueous dunes. *Journal of Geophysical Research: Earth Surface*, 127(2):e2021JF006492.

Jarvis, P. A., Narteau, C., Rozier, O., and Vriend, N. M. (2023). The probabilistic nature of dune collisions in 2D. *Earth Surface Dynamics*, 11:803–815.

Jia, P., Andreotti, B., and Claudin, P. (2017). Giant ripples on comet 67p/churyumov–gerasimenko sculpted by sunset thermal wind. *Proceedings of the National Academy of Sciences of the United States of America*, 114(10):2509–2514.

Katsuki, A., Kikuchi, M., Nishimori, H., Endo, N., and Taniguchi, K. (2011). Cellular model for sand dunes with saltation, avalanche and strong erosion: collisional simulation of barchans. *Earth Surface Processes and Landforms*, 36(3):372–382.

Katsuki, A., Nishimori, H., Endo, N., and Taniguchi, K. (2005). Collision dynamics of two barchan dunes simulated using a simple model. *Journal of the Physical Society of Japan*, 74(2):538–541.

Kroy, K., Sauermann, G., and Herrmann, H. J. (2002). Minimal model for sand dunes. *Phys. Rev. Lett.*, 88:054301.

Lam, D. K., Remmel, T. K., and Drezner, T. D. (2011). Tracking desertification in california using remote sensing: A sand dune encroachment approach. *Remote Sensing*, 3(1):1–13.

Lefebvre, A. and Cisneros, J. (2023). The influence of dune lee side shape on time-averaged velocities and turbulence. *Earth Surf. Dynam.*, 11:575–591.

MANTZ, P. A. (1978). Bedforms produced by fine, cohesionless, granular and flakey sediments under subcritical water flows. *Sedimentology*, 25(1):83–103.

Marvin, M. C., Lapôtre, M. G. A., Gunn, A., Day, M., and Soto, A. (2023). Dune interactions record changes in boundary conditions. *Geology*, 51(10):947–951.

Narteau, C., Zhang, D., Rozier, O., and Claudin, P. (2009). Setting the length and time scales of a cellular automaton dune model from the analysis of superimposed bed forms. *Journal of Geophysical Research*, 114.

Nield, J. M., Wiggs, G. F. S., Baddock, M. C., and Hipondoka, M. H. T. (2017). Coupling leeside grainfall to avalanche characteristics in aeolian dune dynamics. *Geology*, 45(3):271–274.

Omidyeganeh, M., Piomelli, U., Christensen, K., and Best, J. (2013). Large eddy simulation of interacting barchan dunes in a steady, unidirectional flow. *Journal of Geophysical Research: Earth Surface*, 118(4):2089–2104.

Rothman, D. H. and Zaleski, S. (2004). *Lattice-Gas Cellular Automata: Simple Models of Complex Hydrodynamics*. Cambridge University Press, Cambridge.

Rozier, O. and Narteau, C. (2012). A real space cellular automaton laboratory (rescal) to analyze complex geophysical systems. pages 5890–.

Rozier, O. and Narteau, C. (2014a). A real-space cellular automaton laboratory. *Earth Surface Processes and Landforms*, 39(1):98–109.

Rozier, O. and Narteau, C. (2014b). A real-space cellular automaton laboratory. *Earth Surface Processes and Landforms*, 39(1):98–109.

Sauermann, G., Kroy, K., and Herrmann, H. (2001). A continuum saltation model for sand dunes. *Physical Review E*, 64:031305.

Sutton, S. L. F., McKenna Neuman, C., and Nickling, W. (2013a). Avalanche grainflow on a simulated aeolian dune. *Journal of Geophysical Research: Earth Surface*, 118(3):1767–1776.

Sutton, S. L. F., McKenna Neuman, C., and Nickling, W. (2013b). Lee slope sediment processes leading to avalanche initiation on an aeolian dune. *Journal of Geophysical Research: Earth Surface*, 118(3):1754–1766.

Thomas, D. S. G., Knight, M., and Wiggs, G. F. S. (2005). Remobilization of southern african desert dune systems by twenty-first century global warming. *Nature*, 435:1218–1221.

Vermeesch, P. (2011). Solitary wave behavior in sand dunes observed from space. *Geophysical Research Letters*, 38(22).

Vriend, N. M. and Bacik, K. A. (2025). The dynamics of sand dunes. *Annual Review of Fluid Mechanics*, 57:55–81.

Vriend, N. M. and Jarvis, P. A. (2018). Between a ripple and a dune. *Nature Physics*, 14:641–642.

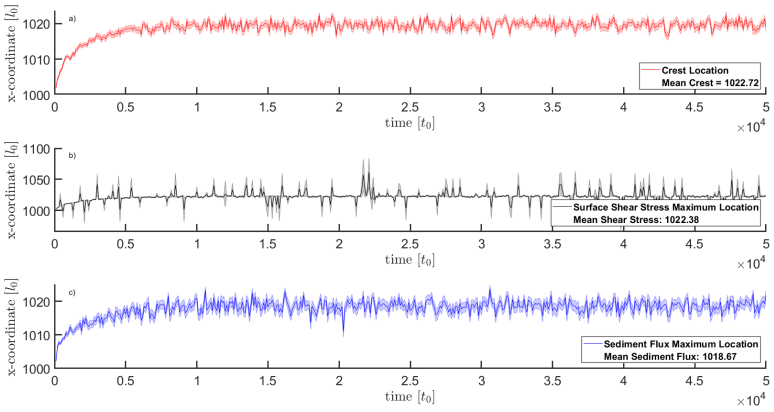
Werner, B. T. and Kocurek, G. (1999). Bedform spacing from defect dynamics. *Geology*, 27(8):727–730.

Zhang, D., Yang, X., Rozier, O., and Narteau, C. (2014). Mean sediment residence time in barchan dunes. *Journal of Geophysical Research: Earth Surface*, 119.

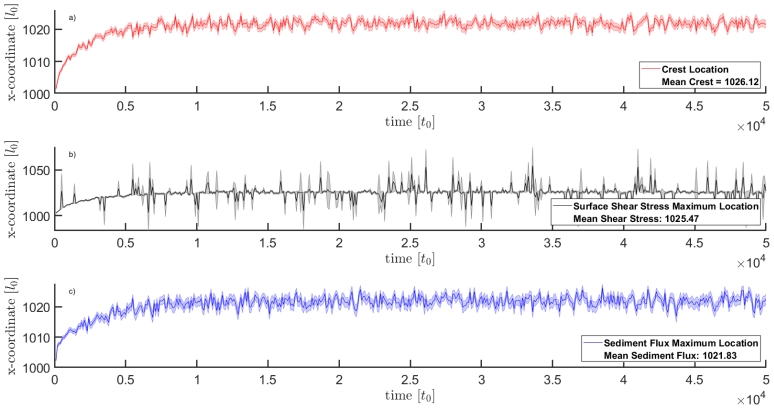
Zhou, X., Wang, Y., and Yang, B. (2019). Three-dimensional numerical simulations of barchan dune interactions in unidirectional flow. *Particulate Science and Technology*, 37(7):835–842.

# 7 Appendix

## 7.1 Additional Standard Error Bars

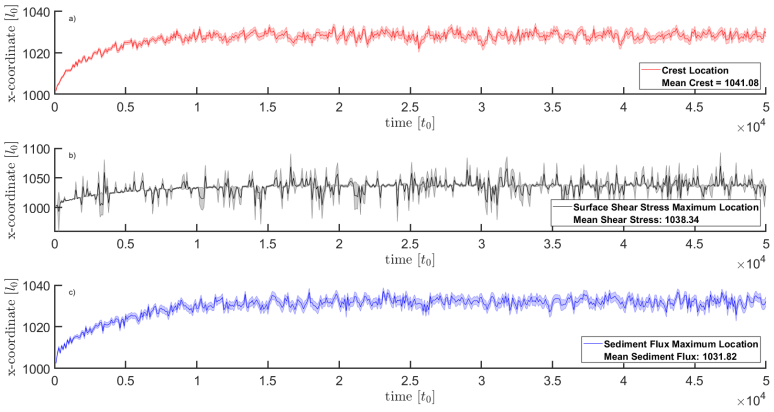



**(a)** $\tau_1 = 1$, with $l_{sat} \approx 3.78\, l_0$

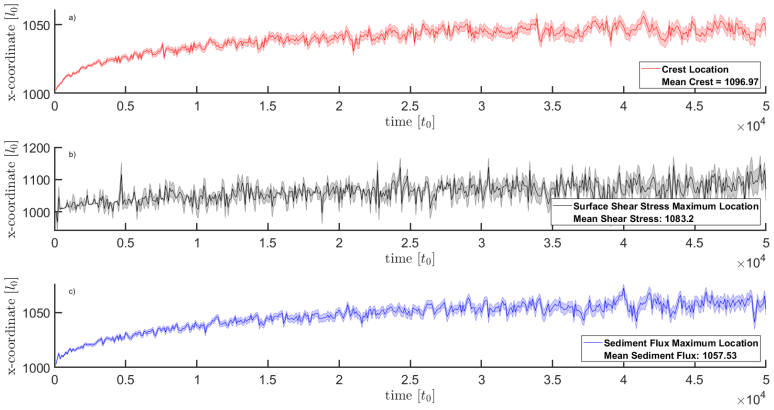



**(b)** $\tau_1 = 50$ with $l_{sat} \approx 3.56\, l_0$

**Figure 29:** Standard Error across fifty simulations. Mean values are calculated from timestep 25,000 onward.




**(a)** $\tau_1 = 150$, with $l_{sat} \approx 4.97\, l_0$




**(b)** $\tau_1 = 300$ with $l_{sat} \approx 26.67\, l_0$

**Figure 30:** Standard Error across fifty simulations. Mean values are calculated from timestep 25,000 onward.